

Product Development Project – MPP126

Redesign of PExA Device for Healthcare Centers



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Abstract

This report presents the redesign of the PExA 2.1 noninvasive respiratory diagnostic device to support its transition from a specialized research tool to a clinical product for healthcare centers. The objective was to improve usability, accessibility, and clinical workflow integration for patients, operators, and service personnel by addressing operational complexities and footprint limitations. Through a structured product development process incorporating pre-studies for user needs analysis, functional decomposition, and matrix evaluations, a final concept based on a compact vertical architecture was selected. Physical prototyping validated critical user interactions, including a motorized lead screw mechanism for precise mouthpiece height adjustment, a digital interface for automated breathing guidance, and outward opening hinged doors for improved service access. While these critical user interaction features were successfully verified, the internal mechanism configuration and structural layout optimization were only explored conceptually, remaining key areas for future physical testing. Alongside the technical development, a commercial assessment evaluated viability, identifying a servitization model within a national screening program as one of the promising pathways to bridge advanced respiratory research with integration into primary care.

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Chapter 1

Introduction

This report is based on a collaboration between Chalmers University of Technology and PExA AB to further develop a non-invasive respiratory diagnostic device. The project focuses on analysing the existing PExA 2.1 device and developing improved prototype concepts using product development methods based on customer needs.

The aim is to move beyond a research focused application and contribute to a more commercialized solution, particularly for lung cancer diagnostics and broader healthcare use.

1.1 Company Background

The origins of PExA AB trace back to research initiated in 2004 by Anna-Carin Olin, a professor at the Department of Occupational and Environmental Medicine at the Sahlgrenska Academy, University of Gothenburg. Her work initially focused on individuals exposed to respiratory hazards in occupational environments, many of whom later developed conditions such as asthma and COPD. Despite clear symptoms linked to workplace exposure, existing diagnostic techniques often failed to reliably detect these diseases. At the time, certain individuals showed elevated levels of exhaled nitric oxide, which indicates lung inflammation associated with allergic asthma. However, this measurement was proved insufficient for diagnosing other forms of asthma or COPD. The recognition of this limited diagnostic capability, prompted Professor Olin and her research team to investigate alternative biomarkers that could be used across a broader range of respiratory conditions. Their efforts at last led to the creation of the first device, the PExA 1.0 [1].

Today, PExA AB is based at Sahlgrenska Science Park in Gothenburg, an innovation hub supporting new health and biotech companies. The company currently offers the PExA 2.1 (Figure 1.1). This is a patented, non-invasive instrument that is used by lung research centers, universities, and pharmaceutical companies to find specific biomarkers that indicates different respiratory diseases [1].



Figure 1.1: PExA 2.1 Machine [1]

1.2 Current Market Position

Today's clinical practice for lung cancer detection is dominated by imaging technologies like LDCT, PET, and MRI, alongside invasive biopsies, which primarily provide analysis only after structural abnormalities are already detectable. This dependence limits diagnosis to later stages and often misses the critical early detection window.

PExA 2.1 enters the market as a complementary breakthrough technology designed to redefine this approach by targeting early biological changes. Functioning similarly to a non-invasive blood test for the lungs, this radiation free method analyzes microscopic particles in exhaled air to detect cellular stress and inflammation originating in the small airways. While PExA 2.1 is currently commercialized as a specialized research instrument, the company's patented biomarker sampling technology still establishes its strategic market position. By identifying lung disease at its earliest biological stage, before tumors become radiologically visible, PExA 2.1 offers a distinct competitive advantage that enables continuous monitoring and significantly earlier treatment interventions [1].

1.3 Product

Currently, the only product offered by PExA AB is the PExA 2.1, a compact instrument measuring 50x50x50 cm and weighing 35 kg, designed primarily for research purposes to sample microscopic particles non-invasively from the deepest regions of the lungs. The device operates through a specific and controlled breathing maneuver. As a patient exhales into a mouthpiece, the instrument captures droplets formed during the natural opening and closing of the alveoli and small airways. These particles include proteins, lipids, and other components of the respiratory tract lining fluid. They are counted, sorted, and collected onto a removable substrate from the impactor for further analysis using established laboratory techniques such as ELISA, mass spectrometry, LC, and HPLC. To ensure standardized and reproducible measurements, integrated software on a computer records the process in real time and tracks breathing patterns, particle counts, total collected mass, and exhaled air volume. This approach provides access to potential biomarkers from the small airways, where many respiratory diseases begin but often remain undetectable by conventional imaging. Overall, the PExA 2.1 improves the understanding of disease mechanisms like lung cancer, offering a platform for earlier detection and supporting the development of more accurate diagnostics and treatments [1].

1.4 Customer

The current customer base for PEXA 2.1 consists of academic and pharmaceutical researchers who use the instrument for non-invasive collection of particles from the small airways. The system uses inertial impaction to enable the study of respiratory diseases at a molecular level. At this stage, the company operates exclusively within the medical research market and does not function as a clinical diagnostic provider. Instead, its primary goal is to support the discovery of new biomarkers for researchers. PEXA achieves this through close customer collaboration, direct technical support, equipment maintenance and repair services, and the continuous supply of disposable consumables [1].

1.5 Resources

Certain key resources for PEXA 2.1 have been identified using the Sustainable Business Model Canvas shown in 1.2 to provide a clear overview of what the company relies on for ongoing product development and daily operations.

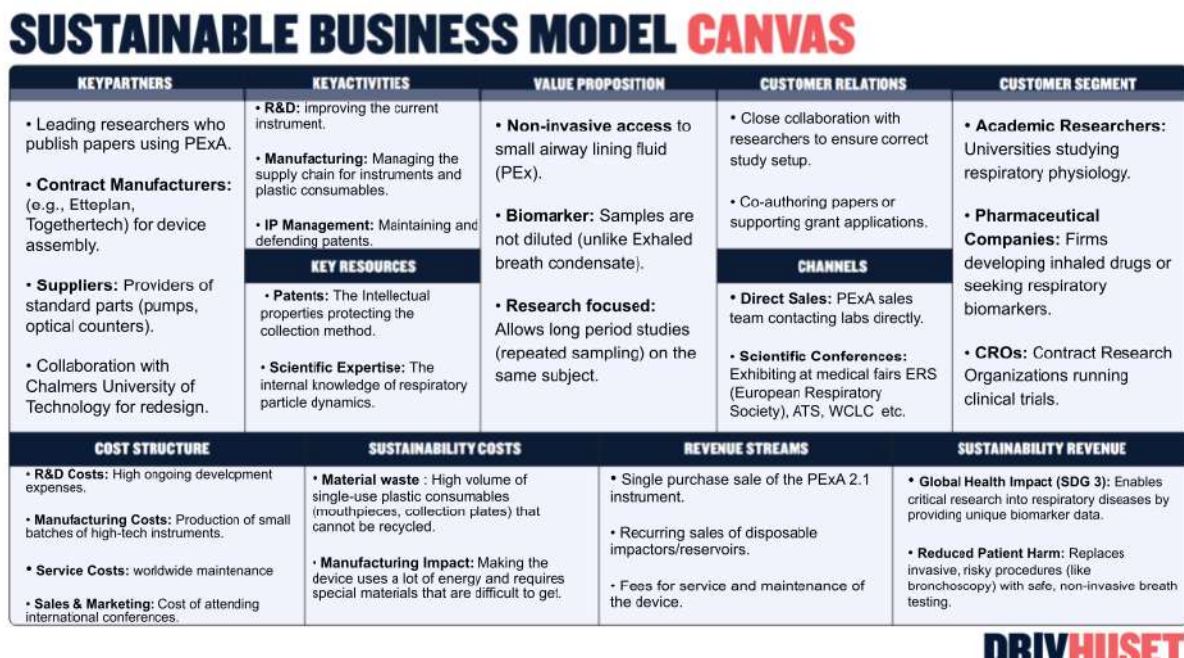


Figure 1.2: Sustainable Business Model Canvas

Intellectual Property and Expertise

PEXA relies on strong scientific expertise and patents that form the core of its value proposition. In-depth knowledge of respiratory particle dynamics, combined with protected collection methods, differentiates the technology from conventional diagnostic tools and enables a non-invasive approach to lung research.

Economy

The company follows a hybrid revenue model, generating income through instrument sales alongside recurring revenue from disposable consumables and maintenance services. This structure supports ongoing development and the production of small-scale high-tech instruments.

Partners

PEXA collaborates with contract manufacturers such as Etteplan and Togethertech for production, and works with specialized suppliers for key components. It also maintains close ties with Chalmers University of Technology and leading academic groups that support product development and validate the technology through research.

Customer Relations

The company serves academic, pharmaceutical, and contract research organizations focused on biomarker discovery. It supports users through close collaboration, technical assistance, and involvement in study design, often contributing to publications and research funding applications.

Social Impact

PExA contributes to global health (UN SDG 3) by enabling research into respiratory diseases and reducing patient risk by replacing invasive procedures such as bronchoscopy with non-invasive breath-based sampling.

Chapter 2

Project Methodology

The project followed a structured product development process, progressing sequentially from user research and requirement specification through concept generation, prototyping, and commercial assessment. Each phase built directly on the outputs of the previous, ensuring that design decisions remained grounded in verified user needs and traceable to defined requirements.

2.1 User Research and Market Pre-Study

A mixed-method exploratory pre-study was conducted to gather qualitative and quantitative data through three primary vectors:

- **Semi-structured interviews** with service technicians and clinical operators to identify workflow, maintenance, and operational bottlenecks.
- **Participatory observation** of the current PExA 2.1 device in operation to isolate ergonomic and usability pain points.
- **Public surveys** to gauge technology acceptance and user readiness regarding the required exhalation maneuver.

This phase successfully identified and segmented critical requirements across three core stakeholder groups: patients (comfort and clarity), operators (simplicity and reduced cognitive load), and service personnel (accessibility and modular maintenance).

2.2 Requirements Specification and Functional Analysis

Qualitative user needs were translated into measurable, testable engineering metrics compiled within a formal Product Requirements Specification Table, categorized into strict Demands (non-negotiable constraints) and Wishes (target optimizations). Concurrently, a system-level functional decomposition was performed utilizing a Means-Function Tree. System actions were expressed in explicit verb-noun pairs to systematically isolate potential technical solution principles for sub-functions, such as airflow control and height adjustment.

2.3 Concept Generation, Screening, and Selection

The conceptual design phase was executed systematically to prevent cognitive bias and explore a wide design space:

1. **Morphological Matrices:** Used to combine distinct solution principles into 13 concepts, developed individually to mitigate groupthink.
2. **Screening Matrix:** Applied to eliminate concepts that violated fundamental geometric boundaries or core functional constraints.
3. **Iterative Pugh Matrices:** Employed to qualitatively evaluate the remaining candidates, using the baseline configuration of the original PExA 2.1 as the initial reference point (*Datum*).
4. **Kesselring Matrix:** Utilized as a weighted scoring framework where explicit weights were assigned to key criteria (usability, serviceability, and manufacturing cost). This provided mathematical justification for selecting the winning *Closet Concept*, which was further optimized by integrating high-performing modular features from runner-up designs.

2.4 Prototyping and Experimental Validation

The project transitioned into a physical, modular prototyping loop to isolate and experimentally validate three critical design upgrades:

- **Lead Screw Mechanism:** to test the vertical adjustment of the patient mouthpiece.
- **Digital User Interface:** Automated instructions for the user guidance via an interactive *Figma* prototype deployed on a digital screen.
- **Enclosure Accessibility:** Built a structural frame using aluminum T-slot extrusion profiles paired with 3D-printed PETG outer panels to evaluate the operational clearance of outward-swinging “closet doors” during rapid internal maintenance.

2.5 Commercial and Exploitation Evaluation

Finally, a comprehensive business viability analysis was conducted to support the future market entry and long-term sustainability of the redesigned device:

- **Regulatory Compliance:** Evaluated alignment with relevant medical device directives, international quality standards, certification frameworks, and target market healthcare regulations.
- **Intellectual Property and Freedom to Operate:** Conducted targeted patent landscape searches and non-infringement assessments to ensure design originality and clear operational freedom.
- **Financial Feasibility and Business Modeling:** Modeled potential market demand based on targeted demographic segments and performed financial projections (such as Net Present Value calculations). This analysis compared a traditional one-time capital purchase business model against an alternative subscription- or service-based model combining hardware acquisition with ongoing consumable contracts.

Chapter 3

Summary of Pre-Studies

This chapter summarizes the findings from the pre-study conducted to support the redesign of the PExA device. The pre-study aimed to establish an understanding of the external market environment, relevant stakeholders, and customer needs associated with transitioning the product from a research instrument to a solution suitable for healthcare centers. The chapter presents the market analysis, identified customer needs, and the resulting design opportunities that formed the foundation for the subsequent product development process.

3.1 Market Analysis

This section examines the current commercial landscape for respiratory diagnostics and early lung cancer screening. It considers present market conditions, identifies the needs of key stakeholders, and reviews existing and alternative solutions in the field. The aim is to highlight current gaps in the market and guide the strategic positioning of the new PExA device within clinical healthcare settings.

3.1.1 Market Characteristics, Trading Obstacles, and Constraints

The broader market characteristics for medical technology and lung cancer screening are shaped by a mix of political, economic, and technological factors. In Sweden, a key defining characteristic is a political landscape driven by national life science and cancer strategies that highly prioritize research-intensive medical technology, prevention, and early detection [2], [3].

Despite this favorable strategic environment, the market faces severe trading obstacles and constraints. From an economic perspective, severe healthcare system pressures act as a primary trading constraint. Rising health expenditures and tight regional budgets caused by inflation [4] force hospitals to prioritize immediate affordability. This economic obstacle delays investments in new premium diagnostic technologies [5].

Furthermore, the PExA device faces major political and legal trading obstacles due to its classification across two strict regulatory frameworks. Because it collects biological samples through direct human contact, it falls under the European Medical Device Regulation (EU-MDR), while its analysis of these samples outside the body categorizes it under the In Vitro Diagnostic Regulation (IVDR). Complying with both sets of rules introduces strict safety and performance constraints [6], [7]. This dual regulation obstacle raises development costs and delays product launches, making the transition from purely research use to full commercialization highly complex, demanding, and currently economically challenging.

Finally, a significant operational constraint within the market is the limited capacity and expertise among medical professionals. The current inability to efficiently interpret new complex biomarker results and manage follow-up care acts as a critical obstacle to widespread clinical adoption [8].

This challenge is further intensified by technological and environmental market demands that act as strict trading constraints for new medical diagnostic devices. From a technological standpoint, the market is shifting away from late-stage hospital diagnoses and demanding early detection capabilities directly within primary care settings. A major operational obstacle for new products is the requirement to connect seamlessly with digital health systems and generate structured biological data designed for predictive AI models [9], [10].

Furthermore, strict public procurement strategies create environmental constraints by legally mandating waste reduction. These environmental trading obstacles mean healthcare systems favor diagnostic approaches that eliminate cumulative radiation exposure and hazardous chemical waste [11].

3.1.2 Target Customers and Stakeholders

Developing a successful diagnostic device requires a clear understanding of both the institutional customers procuring the technology and the end users interacting with it in clinical environments.

Institutional Customers

The primary institutional customers and adoption sites for the device include hospitals, specialized respiratory clinics, and preventive screening centers.

Secondary markets with higher flexibility regarding cost include private health testing facilities. Furthermore, academic researchers, pharmaceutical companies developing inhaled drugs, and Contract Research Organizations (CROs) running clinical trials represent an ongoing established customer base for biomarker research [12].

Primary Users and Operators

In the context of medical device operation, the daily primary users of the physical device will be health center personnel and nursing staff. These operators require equipment that provides clear performing instructions and supportive signaling of a patient's breathing status, since current designs lack indicators for important breathing states, making manual timing unreliable.

Target Patient Segment

The target demographic for the device consists of individuals at a high risk of developing lung cancer. This specific customer segment includes long-term smokers, individuals with occupational exposure to harmful airborne substances, and populations over the age of 55 [13], [14], [15].

While these patients are the clinical focus and act as secondary users of the hardware, their physical experience is directly linked to the success of the operator workflow. The ergonomic design of the mouthpiece and the clarity of the breathing instructions must be optimized for patient comfort and compliance. If a patient struggles to perform the required breathing maneuver, the operator must spend additional time coaching them and repeating procedures, which severely reduces overall clinic efficiency. Therefore, optimizing the physical experience for the patient directly supports the primary user's need for a smooth, reliable, and fast clinical process.

Regulatory Authorities and Legal Requirements

To guarantee that the device meets all medical regulations for safe use in Healthcare centers, the system shall comply with specific regulations:

- Regulation (EU) 2017/745 (MDR): To meet strict legal safety and performance requirements for medical devices in the EU [6].
- ISO 13485: To establish a comprehensive quality management system that ensures controlled processes, strict regulatory compliance, and strong patient safety [16].
- ISO 14971: To systematically identify, evaluate, and control associated risks throughout the entire lifecycle of the medical device [17].
- IEC 60601-1: To guarantee the basic electrical safety, performance, and electromagnetic compatibility (EMC) of the equipment in a clinical setting [18].

- Furthermore, the collection and handling of patient data must strictly comply with GDPR to ensure ethical protection [19].
- Device interoperability will also benefit from alignment with upcoming European Health Data Space (EHDS) regulations [20].

3.1.3 Competitive Landscape and Alternative Solutions

Currently, the device does not have a direct competitor offering an identical detection principle or clinical role. Instead, it operates within a landscape of alternative lung assessment technologies that evaluate different diagnostic aspects [21]. This unique position allows the technology to function as a complementary pre-screening solution that supports existing lung cancer screening workflows.

3.1.3.1 Evaluation of Competing Products

Low Dose Computed Tomography (LDCT)

LDCT is the primary imaging technology currently used for structural lung cancer screening [22]. It is the most widely adopted method for detecting early signs of lung cancer and remains the only screening approach proven to reduce mortality rates [23]. However, LDCT is a highly resource-intensive structural imaging tool that requires expensive equipment and specialized personnel for interpretation. It also exposes patients to cumulative radiation and cannot detect early biochemical changes. Although clinical demand for LDCT screening is high, implementation is often constrained by scanner availability, healthcare resources, and the organizational capacity needed to manage large-scale screening programs [24].

Spirometry and FeNO Analyzers

Spirometry and FeNO analyzers represent the most established alternative diagnostic tools for respiratory assessment. These systems are relatively low-cost, widely accessible, and available in multiple device variants, resulting in extensive use across global healthcare settings. However, they primarily evaluate airflow limitation, airway mechanics, or inflammatory status through single gaseous biomarkers [25]. As a result, they provide limited functional insight and lack the ability to detect early molecular changes associated with lung cancer [26].

3.1.3.2 Strategic Positioning and Identified Market Gaps

As demonstrated by the limitations of these established methods, a clear clinical gap exists. Most existing tools either capture late-stage structural abnormalities once a tumor is visible or provide basic lung function measurements. The new device is positioned as a complementary non-invasive technology that fills the biochemical diagnostic gap. It enables radiation-free early-stage risk assessment from the distal small airways and alveoli. This approach successfully avoids the operational constraints and patient risks associated with LDCT imaging or invasive biopsies [27], [28].

3.1.3.3 Supply Chain and Market Channels

To successfully deliver this complementary technology to the clinical market, a reliable manufacturing and distribution infrastructure is required. The supply chain for advanced diagnostic instruments such as the PExA device relies on a network of specialized partners and subcontractors. This network includes contract manufacturers for device assembly alongside external engineering partners who provide prototyping expertise [29]. Furthermore, the manufacturing process depends heavily on providers of standard mechanical components such as vacuum pumps and highly specialized optical particle counters [30].

Market channels consist primarily of direct sales driven by both inbound inquiries and targeted outreach to clinical laboratories and research departments [29]. Active exhibitions at major scientific medical conferences such as the

European Respiratory Society, the American Thoracic Society, and the World Conference on Lung Cancer act as vital marketing instruments alongside traditional advertising and direct distributor networks [31].

3.1.4 Concluding Market Requirements

Based on the preceding market analysis and competitive evaluation, the following key requirements presented in Table 3.1 must be fulfilled for the device to achieve successful commercial adoption in healthcare centers:

Requirement Category	Specific Market Requirements
Noninvasive Safety	• Must be radiation-free and non-invasive. • Aligns with regional sustainable procurement strategies aimed at reducing hazardous medical waste [11].
Digital Integration	• Must generate structured biological data for secondary AI applications. • Must align with GDPR and upcoming European Health Data Space (EHDS) regulations [19], [20].
Operational Suitability	• Must be operable by primary care nursing staff without the need for specialized hospital infrastructure. • Requires clear visual or physical indicators for patient breath states to prevent manual timing errors.
Economical Footprint	• Must feature a slim and tall design to fit on limited desk space within standard health centers. • Must utilize standard components where possible to reduce the high cost per test and fit within tight regional healthcare budgets [4].
Regulatory Compliance	• Must meet strict standards under the EU Medical Device Regulation (MDR) and In Vitro Diagnostic Regulation (IVDR [6], [7]).

Table 3.1: Concluding Market Requirements for Successful Commercial Adoption

3.2 Customer Needs

To better understand the requirements for transitioning the PExA device from a research instrument to a product suitable for healthcare centers, a customer needs analysis was conducted. Information was gathered through stakeholder interviews, participatory observation of the current device, and a questionnaire targeting potential future users. The collected data were analyzed and organized into prioritized customer needs.

Patients expressed a need for a system that is intuitive, comfortable, and easy to understand without extensive guidance. The breathing maneuver required by the current device was perceived as complex, highlighting the importance of clear instructions and supportive feedback throughout the procedure. Accessibility and confidence during use were identified as important factors for improving the overall patient experience.

Healthcare personnel emphasized the need for a solution that minimizes workload and reduces the amount of supervision required during testing. In busy healthcare environments, efficient workflows and short training times are essential. Participants therefore highlighted the importance of clear system feedback, simplified operation, and features that support more independent patient use.

Service personnel focused primarily on maintenance and accessibility. The current device contains several components that are difficult to access during servicing, increasing both maintenance time and complexity. Easy access to internal components, modular design solutions, and straightforward cleaning procedures were identified as important requirements for future development.

Across all stakeholder groups, a common theme emerged: the need to transition the PExA device from a research-oriented instrument operated by specialists to a more accessible and user-centered product suitable for routine

clinical use. These identified customer needs formed the foundation for the requirement specification and guided the subsequent concept development process.

3.3 Description of the need for a new product

The need for a new product is primarily driven by a shift in the intended context of use. The current solution is designed for research environments, where trained personnel operate the machine under controlled conditions. However, there is an increasing demand to move the product to healthcare centers, where it can be used more frequently and by a broader range of users. This transition introduces new requirements regarding usability, accessibility, and integration into clinical workflows. Healthcare environments are often time-constrained and require efficient processes, meaning that solutions must be intuitive and easy to operate. Therefore, improving accessibility and ease of use is essential to ensure that the product can be adopted in everyday clinical practice and contribute to more efficient healthcare delivery.

3.3.1 Limitations of the Current Solution

The current product is not fully adapted for use in healthcare centers. Its design assumes the presence of trained personnel and controlled usage conditions, which limit its applicability in more general clinical settings. Key limitations include operational complexity, dependence on assistance from healthcare personnel, and challenges related to integration into everyday clinical routines. These factors can lead to inefficiencies in healthcare workflows and an increased workload for staff. Furthermore, the interaction between the patient and the machine is not optimized for independent use, reducing accessibility and limiting the potential for broader implementation. As a result, the current solution does not fully support the needs of a more scalable and patient-centered healthcare context.

These limitations point toward an open-ended design problem. One potential direction is to enable patients to manage the entire process independently, thereby reducing the workload on healthcare personnel. However, such a solution would require the machine to be highly intuitive, allowing first-time users to understand the procedure with minimal guidance. This could be achieved through clear instructions and supportive system features, such as automation, voice guidance, and other interactive elements that facilitate user-machine interaction. An alternative approach is to maintain the involvement of healthcare personnel but improve the efficiency of the interaction between the user and the machine. In this scenario, the focus would be on simplifying operations, reducing complexity, and optimizing workflows so that staff can assist patients more quickly and with less effort. This could involve improved interface design, clearer feedback from the system, and better integration into existing clinical routines.

These two directions represent different strategies for addressing the identified limitations, highlighting a trade-off between increased patient autonomy and continued reliance on healthcare personnel. This reinforces the need to further explore how the product should balance usability, efficiency, and responsibility in a healthcare context.

3.3.2 User Interaction and Stakeholder Needs

The product involves, as mentioned before, multiple user groups, including patients, healthcare personnel, and service personnel, each with different needs and expectations.

From a user-centered design perspective, it is essential to consider the interaction between all stakeholders and the machine as part of a larger system. Patients require a solution that is intuitive, safe, and easy to understand, minimizing uncertainty during use. Healthcare personnel need a system that reduces workload and allows for efficient operation without extensive training, as well as a machine that is easy to clean, as it is placed in sterilized environments. Service personnel require a design that simplifies maintenance, troubleshooting, and system handling.

The findings indicate that improvements must be made not only to individual interactions but to the overall system of use. A holistic approach to user interaction is therefore necessary to ensure that the product functions effectively

in real-world healthcare environments.

3.3.3 Opportunity for Increased Autonomy and Accessibility

An important opportunity identified in the project is to enable more independent use of the product. There is a potential need for a solution where patients can perform the procedure on their own, without direct assistance from healthcare personnel. Although this direction is not yet fully defined, it reflects a broader trend toward increased autonomy and accessibility in healthcare. Enabling independent use could significantly improve efficiency by reducing the need for staff involvement, while also increasing the availability of the procedure to a larger number of patients. This shift toward greater autonomy aligns with the development of more patient-centered healthcare systems, where solutions are designed to empower users and improve overall accessibility.

3.3.4 Design Opportunity and Vision

Based on the identified needs and limitations, there is a clear opportunity to develop a product that is better adapted for healthcare environments. The new solution should support the transition from a specialized, expert-driven research tool to a general-use product suitable for clinical settings. By applying principles of user-centered design, the product can improve usability, reduce complexity, and enhance interaction across all stakeholders. Additionally, by focusing on efficiency and accessibility, the solution has the potential to streamline healthcare workflows and contribute to more effective resource utilization.

Ultimately, the design opportunity lies in developing a product that not only improves functionality but also responds to broader trends in healthcare, including increased accessibility, patient autonomy, and the need for efficient, scalable solutions.

Chapter 4

Concept Development Results

This chapter presents the outcomes of the concept development phase. Guided by the requirement specification established from the pre-study, multiple design concepts were generated and refined through an iterative development process. The concepts were systematically evaluated using structured selection methods to identify the solution that best fulfilled the project requirements. The chapter also presents the selected concept and the key design decisions that formed the basis for the prototype development.

4.1 Requirement Specifications and Early Prototyping

The development process was guided by a structured requirement specification, which served as a central framework for decision-making throughout the project. Building upon the initial specification and identified needs developed during the pre-study, the requirements were refined and extended to support engineering development and concept evaluation.

The initial specification primarily captured customer needs and qualitative expectations. However, to enable systematic product development, these were translated into measurable and testable requirements. This included defining performance metrics and corresponding target values, allowing concepts to be evaluated based on objective criteria rather than subjective judgment.

A part of the final requirement specification is presented in Figure ??, see full version in Appendix A. Compared to the initial version, several requirements were refined with clearer target values and improved prioritization. This increased the usability of the specification as a decision-support tool during concept selection and development.

REQUIREMENT SPECIFICATION		Issued by: Project group PExA				
Category	Requirement	R/W	Importance	Values	Confidence with the value (1-5)	Justification (why it is a requirement)
Functional Requirement	Adjustable Mouthpiece Height The mouthpiece height shall be adjustable within a vertical range of 0-30 cm.	R		0-30 cm	4	To ensure users of all sizes can sit comfortably and reach the mouthpiece easily.
	Airflow Resistance Reduction The airflow resistance through the mouthpiece and filter system shall be at least 15% lower compared to the reference design	W	10	15 %	1	To reduce the effort needed for a patient to breathe through the filter system.
	Sensory Feedback Recognition At least 90% of users shall correctly recognize and interpret the sensory feedback provided during operation.	R		>90%	4	To ensure most users can finish every step correctly on their own.
	Valve Operation The Valve shall be operable with a single action and require a maximum force of 1.5 N	R		≤ 1.5 N	3	To ensure the valve is easy to operate with a single action without straining the hand.
	Proper Installation Indication The system shall provide a clear mouthpiece installation indicator that is correctly recognized by the operator.	W	4	>90%	4	To ensure most operator scan install the mouthpiece correctly and safely without help.
	Internal Component Accessibility Components requiring maintenance shall be accessible within no more than three disassembly steps	R		2-3 steps	3	To minimize downtime and ensure easy maintenance by service personnel.
	Leakage Prevention The system shall prevent air leakage under normal operating conditions.	W	11		1	Sometime the machine leaks which is not sufficient
	Breathing Pause Capability The system shall allow the patient to pause between breathing cycles for 5-10 seconds without affecting	W	9	5-10 sec	3	To give patients a moment to relax between breaths without stopping the test
Usability Requirement	Learnability A first-time user shall be able to understand and complete the full operating procedure within 1-3 minutes, without external assistance	W	5	1-3 min	3	Easier to understand the procedure
	Mouthpiece Comfort The mouthpiece shall enable users to comfortably maintain it in the mouth for at least 30% longer duration compared to the previous design	W	8	30%	2	To ensure comfort during extended periods of use
	Mobility The system shall be movable by one person without mechanical lifting assistance	W	1	1 person	3	To ensure the system can be moved safely by one person between health center rooms.
	Cleanability All external surfaces shall be cleanable within 2 minutes, following the defined cleaning procedure.	R		2 mins	2	To ensure the device is safely decontaminated between users in a short timeframe.
	Assembly Time The system shall be assembled within 5 minutes, using step-by-step assembly instructions	W	7	5 mins	2	In order to stream line the assembly and minimize lead times

Figure 4.1: Requirement Specification Table

As the project progressed, the requirement specification was iteratively updated to reflect new insights gained during concept development and early prototyping. Trade-offs between conflicting requirements were continuously considered, particularly in areas related to usability, technical feasibility, and system performance. This ensured that the specification remained aligned with both user needs and practical constraints.

In parallel with the refinement of requirements, early experimental prototyping was conducted to explore and evaluate initial solution ideas. These prototypes, see figure 4.2, were intentionally simple and focused on testing specific functions rather than representing complete product concepts. By quickly building and assessing physical representations, the team was able to identify promising solution directions and eliminate less feasible alternatives at an early stage. The early prototypes also supported communication within the team by making abstract ideas more tangible. This facilitated discussion, feedback, and iterative improvement of concepts during the initial phases of development.

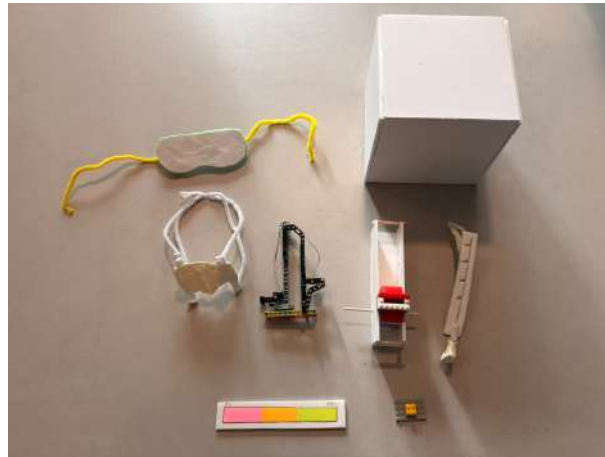


Figure 4.2: Result of Experimental Prototyping

4.1.1 Core Design Requirements and Specifications

Building on the requirement specification shown in ??, this section highlights the main design constraints guiding the concept development. It summarizes the core requirements the new device must meet to perform successfully. By balancing market demands with technical capabilities and sustainability goals, these points set the primary boundaries for the project.

Customer and Usability Requirements

Based on the previous customer needs study, the device must be highly intuitive for first-time patients in terms of both physical interaction with the mouthpiece and understanding of the breathing maneuver instructions. This intuitive design improves the overall sampling process and simplifies the experience for both the patient and the operator.

Furthermore, the system must require minimal training for primary care operators. This ease of use is critical to avoid placing additional strain on healthcare facilities that already experience significant personnel shortages. At the same time, to support service personnel, all internal components must be easily accessible for routine maintenance and repair.

Technical and Geometric Requirements

Technical constraints require a strict reduction in physical volume. Specifically, the total system volume must be less than 0.125 cubic meters compared to the original dimensions, while maintaining a form factor that fits within limited desk space in standard health centers. Additionally, the device must operate using standard electrical outlets and produce low noise levels to maintain a calm clinical environment.

Environmental and Societal Sustainability

Societal sustainability targets focus on increasing accessibility to early lung cancer screening for high risk demo-

graphics [13], [14]. Environmental requirements dictate the transition to recycled plastics for the external casing. It is important though to distinguish the cases in which recycled plastics can be used as the contamination risk and property variability makes recycled plastics unsuitable for the case. Utilizing plastic instead of the previous sheet metal significantly improves thermal insulation. This superior insulation holds the internal temperature for a longer duration which reduces the energy consumed during the sample heating phase and decreases the overall starting time of the machine.

Economical Requirements

Finally economical requirements demand a drastically lower cost per sample to ensure the device fits within tight regional healthcare budgets [4]. This vital cost reduction will be achieved over time as the diagnostic focus narrows from an expensive broad biomarker analysis down to a targeted specific analysis once the exact particles associated with specific respiratory diseases are identified [27].

4.2 Concept Generation

The concept development phase was structured to systematically generate and refine design solutions based on the defined requirement specification. The process combined functional analysis, structured idea generation, and iterative concept development to ensure both breadth and feasibility in the generated concepts.

Initially, a functional decomposition of the system was conducted using a means-function approach, see figure 4.3. This resulted in a set of sub-functions, which were used as a foundation for idea generation. The overall function of the system was broken down into a set of sub-functions, expressed using verb-noun formulations. This enabled a structured understanding of how the product operates and clarified the relationships between different functional elements. The resulting function structure provided a foundation for identifying solution principles for each sub-function.

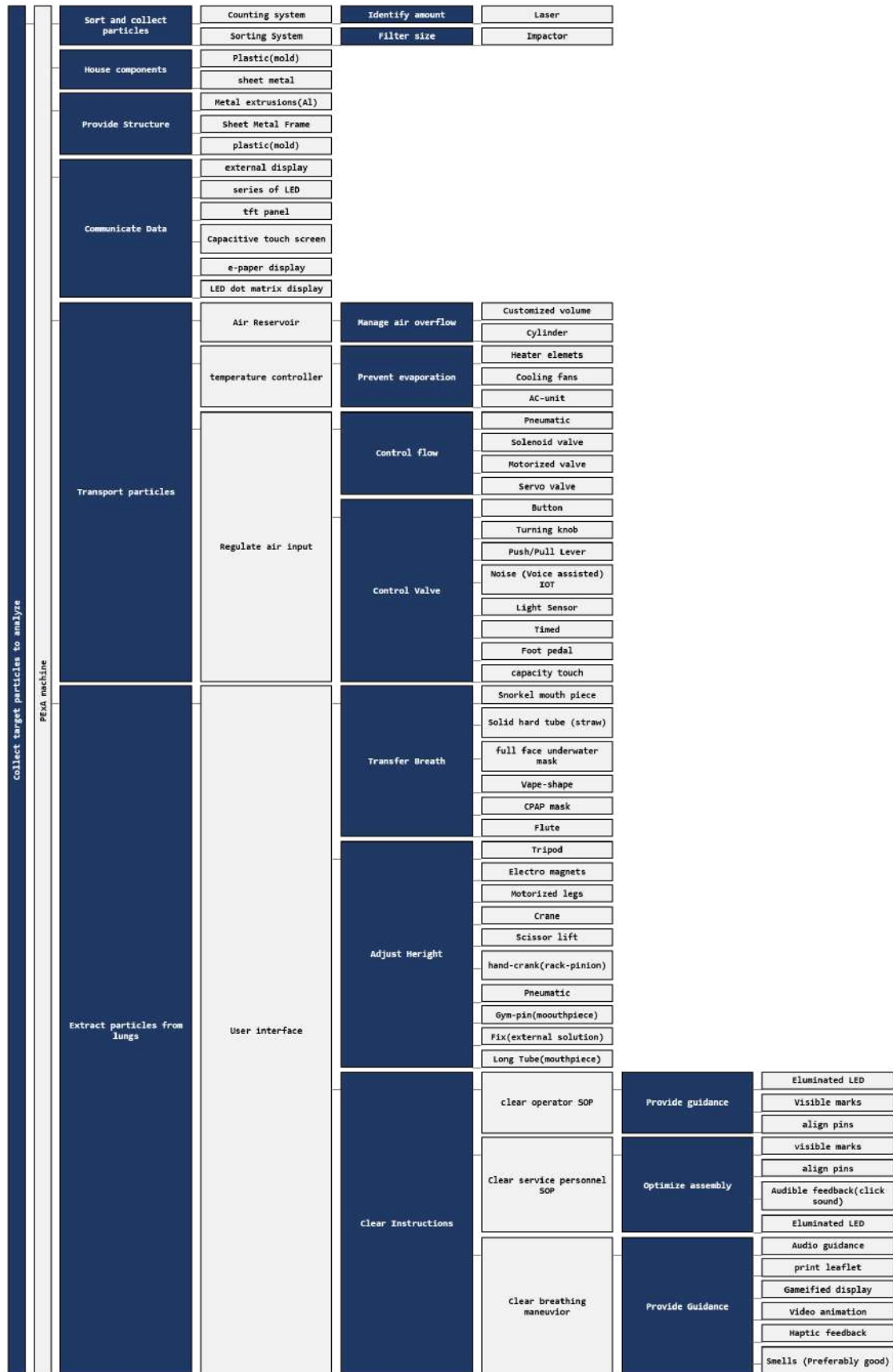


Figure 4.3: Mean-Function Tree

Based on these sub-functions, a first morphological matrix, see figure 4.4 was developed, where each function was associated with a set of alternative solution principles.

Sub-functions	Sub-Solutions									
Adjust height	Tripod	Electro magnet	Motorized Legs	Crane	Scissor Lift	Hand-crank	Pneumatic	Gym-pin	Long tube	Fixed
Breath Input	Snorkel-type mask	Solid Hard Tube (Straw)	Vape-shape	CPAP Mask						
Communication usage steps	Audio Guidance	Print Leaflet	Video Animation	Haptic Feedback						
Control Airflow	Pneuma-tic	Electro Mechanical Actuation	Linear Motion							
Control Valve	Button	Lever	Voice Actovated	Light Sensor	Times					
Visulize Data	Internal Display	External Display								

Figure 4.4: First Morphological Matrix

The initial matrix enabled structured idea generation, where compatible sub-solutions were combined into complete concepts. To generate these sub-solutions, brainstorming-based ideation sessions were conducted, where the team collaboratively explored multiple solution alternatives for each sub-function. During these sessions, evaluation of ideas was deliberately postponed to encourage a broad and unconstrained exploration of the solution space. Through this process, a total of 13 initial design concepts were generated, representing different approaches to fulfilling the identified functions.

To reduce the number of concepts and focus the development effort, an early concept screening was performed using an elimination matrix. The evaluation criteria were derived from key requirements and included factors such as vertical adjustability, sensory feedback, valve operation, compact volume, and cost control. Concepts that did not satisfy these criteria were removed from further consideration. This process resulted in the elimination of four concepts, while the remaining concepts fulfilled the basic requirements but required further development to become complete design solutions. Following this initial screening, the concept development process was continued through an iterative refinement approach. The morphological matrix was expanded to include additional functions necessary for a more complete system representation, see figure 4.5. This second iteration was supported by further brainstorming and idea development, allowing new solution principles to be generated and integrated into the design space.

Sub-functions	Sub-Solutions									
Adjust height	Tripod	Electro magnet	Motorized Legs	Crane	Scissor Lift	Pneumatic	Gym-pin	Long Tube	Fixed	Lead Screw
Transfer Breath	Snorkel-type mask	Solid Hard Tube (Straw)	Vape-shape	CPAP Mask	Current					
Provide Guidance (Patient)	Audio Guidance	Print Leaflet	Video Animation	Haptic Feedback	Smell					
Provide Guidance (Operator)	Eluminated LED	Visible Marks	Align Pins							
Provide Guidance (Assembly)	Visible Marks	Alignment Pins	Pull out drawer (dishwasher)							
Control Airflow	Pneuma-tic	Electro Mechanical	Linear Motion	Mechanical						
Control Valve	Button	Lever	Voice Activated	Light Sensor	Foot Pedal	Capacity Touch				
Store Airflow	Customized	Current solution (rectangle)	Cylinder							
Maintaine Temperature	Heater Element (Current)	Cooling Fans	Ventilation Holes	AC unit						
Communicate data	External Display	Series of LEDS	TFT PANEL	E paper Display	LED dot matrix	Capacity touch screen				
Provide Structure (Internal)	Aluminum Extrusions	Sheet Metal frame	Plastic	Wood						
House Components (External)	Plastic Mold	3D printed	Sheet Metal							

Figure 4.5: Second Morphological Matrix

Using the updated matrix, new combinations of sub-solutions were generated, resulting in six more developed and complete design concepts. These concepts incorporated a broader set of functional requirements and provided a stronger basis for further evaluation. To further support diversity and avoid early convergence, the concepts were developed individually by each team member. This approach encouraged independent thinking and resulted in varied design proposals with different technical and functional characteristics. The individually developed concepts were then presented and discussed within the group in a structured session, where feedback was provided, and ideas were further refined. The concepts are presented below in Chapter 5. This iterative concept development process, combining structured methods with brainstorming-based ideation, enabled a transition from a broad set of initial ideas to a smaller number of more refined and technically feasible design concepts, forming a strong foundation for the subsequent concept selection phase.

4.3 Proposed Design Concepts

The concept development of the new PExA device builds directly upon the morphological matrix functions derived from the requirement specification table established in Chapter 4. Utilizing the second morphological matrix fig. 4.5 to define twelve critical device functions such as mouthpiece height adjustment and user guidance supported the systematic generation of concepts by combining them with various alternative solutions.

To create a comprehensive design, the idea generation phase was widened to consider the product as a whole rather than focusing exclusively on individual functions. Concept ideas were initially developed independently to ensure autonomous thinking and avoid early group bias. This specific strategy encourages diverse perspectives and leads to more thoroughly developed and varied ideas. These conceptual redesigns integrate essential requirements and desirable features to produce viable technical solutions supporting the transition of the PExA device from a research focused piece of equipment into a commercial clinical instrument [12].

Using this independent idea development strategy, the resulting concepts were visualized in several different ways. Some concepts were developed using CAD and rendered visualizations to provide a clearer spatial and structural understanding, while others utilized detailed sketches to highlight the internal and external functionalities of each product concept.

The main goal of the idea generation process was to create a balanced product focused on the three primary user groups: the patient, the operator, and the service personnel. The design process attempts to resolve critical technical problems, including the large physical footprint, limited internal component space for servicing, mouthpiece comfort, and the need for clear instructions due to the complex breathing maneuver.

Technical verification of these critical systems was performed through prototype testing detailed in Chapter 6, alongside usability evaluations. In total, six distinct and detailed concepts were developed to address these technical bottlenecks and limitations from multiple perspectives. These six concepts are presented below:

4.3.1 The Dishwasher concept

Characterized by a quadrant shaped design with rounded edges, the Dishwasher concept creates a more compact form compared to the original device, see Figure 4.6. The structure is built using aluminum extrusions to provide stability and structural strength. Patient height adjustment is achieved through a gym pin mechanism that enables simple and fast positioning changes. Airflow control is managed using a push button that opens and closes the internal valve using an electric valve actuator during the breathing maneuver.

Patient guidance is supported through printed breathing instructions, while the mouthpiece is connected to a long flexible tube that allows flexible height adjustment and improved ergonomic positioning. A central feature of this concept is the drawer system inspired by a dishwasher tray mounted on sliding rails. All major internal components are integrated into the drawer, providing easier access during maintenance and servicing. This design specifically addresses problems identified during semi structured interviews with service personnel, where the current device was described as difficult to service due to the limited space between components, leading to complicated access procedures and many steps during maintenance.

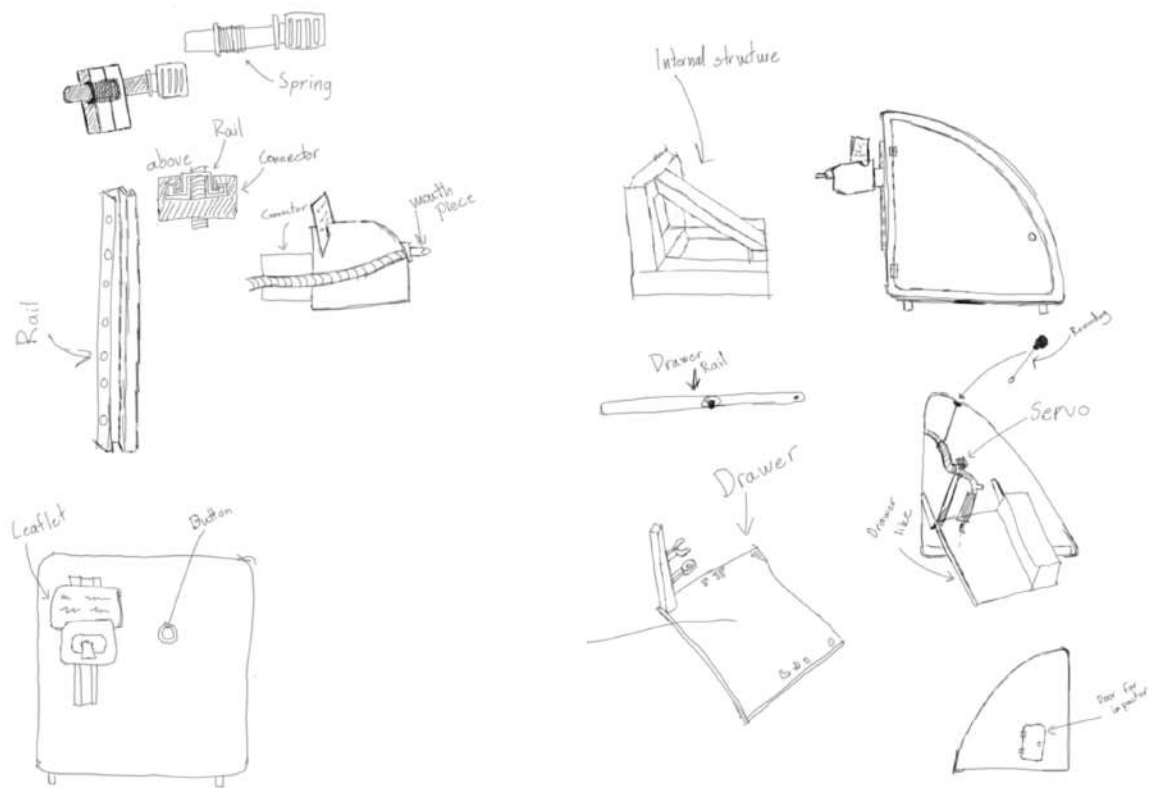


Figure 4.6: The Dishwasher Concept

4.3.2 The Baba Yaga concept

Emphasizing automation and digital integration, the Baba Yaga concept achieves mouthpiece height adjustment through motorized linear actuated legs, see Figure 4.7. These legs raise or lower the entire unit, eliminating the need for a separate height adjustable table. A large integrated screen positioned at the center of the device provides visual guidance for both the patient and the operator during the breath sampling procedure.

The patient breathes into a mouthpiece connected to a soft retractable tube that can be extended or retracted into the machine depending on the patient's preference and ergonomic needs. This concept focuses on improving usability, flexibility, and user interaction through automated adjustment systems and integrated digital guidance.

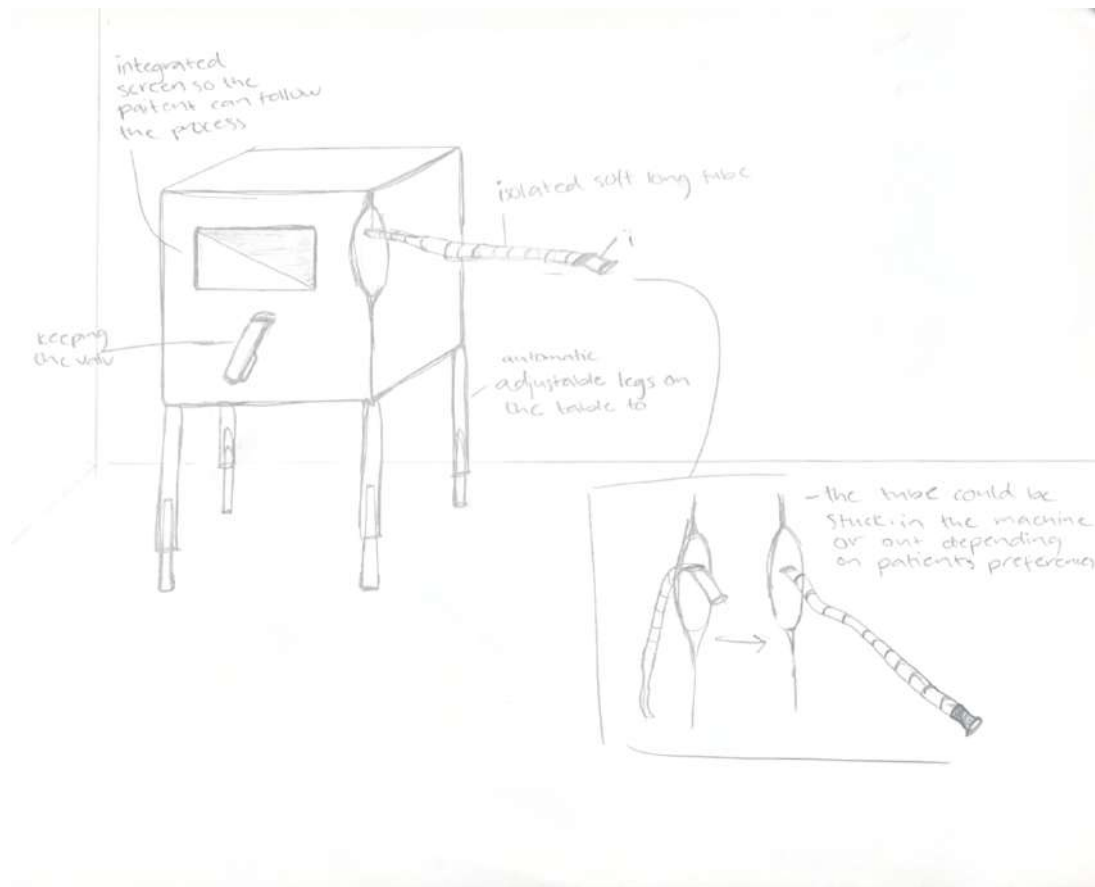


Figure 4.7: The Baba Yaga concept

4.3.3 The Leadacor concept

The Leadacor concept utilizes a taller in height compared to the existing PExA machine and employs a stepper motor combined with a lead screw mechanism to adjust the height of the disposable mouthpiece, see Figure 4.8, which also securely locks its position due to friction. The mechanism is housed inside the machine, which is constructed using an aluminum extrusion frame and removable injection molded plastic panels. These panels are coated with EMC tape to shield the electronic components. A steel bellows cover protects the internal parts from dust and external elements.

The need of "off the shelf" parts utilizes the existing mouthpiece but rotates it to a 45 degree angle facing up the mouth of the patient, improving effectively patient comfort. The current mechanical lever for airflow control is replaced with an electromechanical actuator controlled by a lever switch. For patient guidance, an LED array and a speaker provide feedback during each breathing cycle, while an integrated LCD display shows the overall progress of the test. A HEPA filter storage compartment is located at the bottom of the machine. Finally, the substrate collection area can be accessed through a side mounted door.



Figure 4.8: The Leadacor concept

4.3.4 The Minimalist concept

Optimizing internal volume, the Minimalist concept takes a cube shaped approach with smaller dimensions of approximately 40 by 40 by 40 centimeters, see Figure 4.9. It rearranges the components by specifically aligning the reservoir tubes more efficiently within the structure and using a cylindrical shaped air storage unit to reduce space requirements. The mouthpiece height is adjusted using a manually adjusted lead screw assembly located on top of the instrument. This manual operation allows for precise locking and easy reversible directions while reducing cost, overall weight, and the risk of motor failure. The top section also includes integrated storage for the HEPA filters, concealed beneath a lockable cover to maintain the minimalist design. User guidance is provided through both a printed leaflet positioned beside the mouthpiece and an integrated display located on top of the instrument.

The mechanical lever is replaced with an electromechanical actuator, which is controlled by a turning knob on the side of the machine. This knob includes a breath and stop indicator similar to a standard lock interface to assist the operator during the procedure. The structure is built using aluminum extrusions featuring low profile inward facing half moon handles on the sides to help with carrying the device while maintaining an easily cleaned smooth surface. Injection molded removable plastic panels fitted with EMC tape are used to reduce overall weight and protect electrical components. Additionally, valve and mouthpiece indicators are placed on the outside tube to ensure easy and correct installation of the three way mouthpiece.

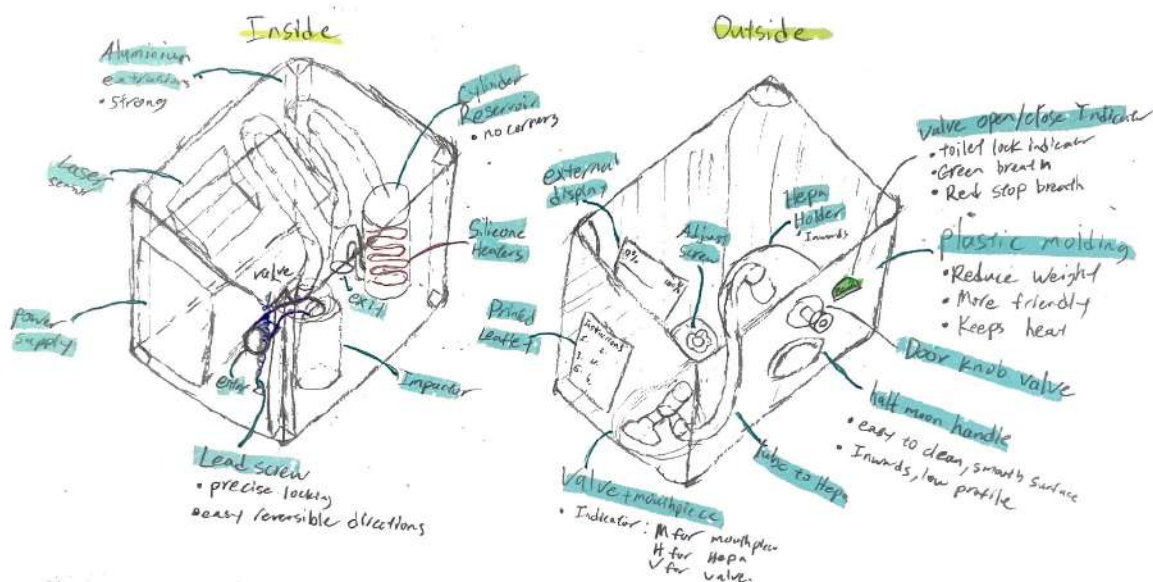


Figure 4.9: The Minimalist concept

4.3.5 The Closet concept

Built upon a tall, vertical rounded rectangular prism structure, the Closet concept utilizes an extendable tube placed outside the machine to facilitate height adjustment, see Figure 4.10. The core structure is built using three aluminum extrusion pillars, creating a taller overall design. This increased height optimizes the internal space and allows easy access to internal components through two hinged plastic doors. These doors open outward to provide smooth and free entry to the interior, making it easy to replace parts or clean the machine when needed.

Airflow is controlled using an electromechanical actuator operated by a linear sliding lever located on the front panel. Because this lever faces the patient, it uniquely allows the user to control the tempo of the breathing maneuver, unlike the other proposed designs. This independent control is quite helpful for experienced patients, enabling them to pace themselves rather than relying entirely on an operator for every step. Furthermore, patient guidance is provided through a foldable LCD screen situated on top of the device displaying clear animated instructions such as when to inhale, hold the breath, and exhale.



Figure 4.10: The Closet concept

4.3.6 The Compact concept

Representing the most space-efficient design among all proposed alternatives, the Compact concept incorporates a triangular shape and relies heavily on a custom printed circuit board (PCB), see Figure 4.11. Implementing a custom electronic board provides several key engineering advantages for this concept. Primarily, it reduces the internal volume required for electronics by integrating multiple control functions that would otherwise require separate standard modules and extensive wiring harnesses. This consolidation not only saves valuable internal space but also improves overall system reliability by reducing the number of physical connection points that could fail during daily clinical use. Furthermore, a specifically designed board allows for optimized electromagnetic shielding, which is crucial for meeting strict medical safety regulations.

To further minimize the overall device footprint, this concept shifts certain user interfaces externally. For breath input, it utilizes a long extendable tube located outside the machine. Interaction and data visualization are handled entirely by an external display monitor paired with a computer mouse. Finally, the internal airflow valve is digitally controlled directly through the custom board, ensuring precise operation without the need for a physical mechanical lever.

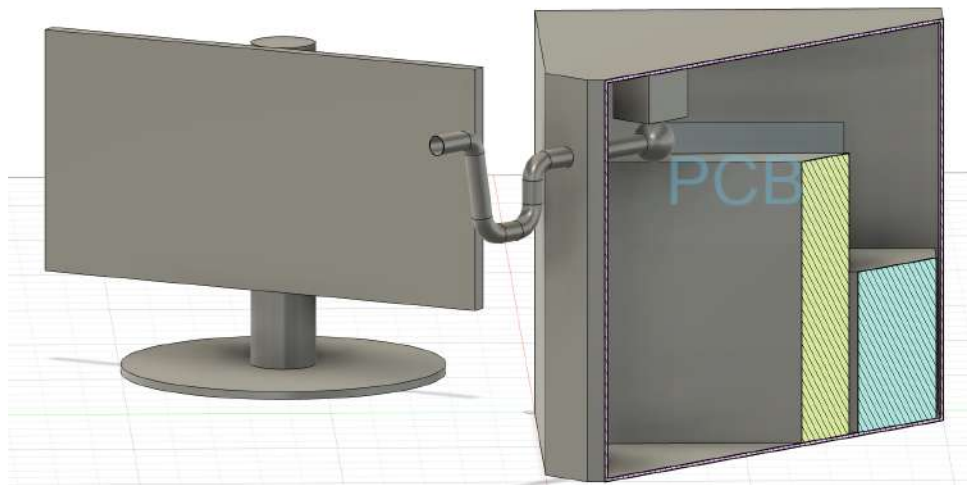


Figure 4.11: The Compact concept

4.4 Concept Evaluation and Selection Process

Following the concept development phase, a structured selection process was carried out to systematically evaluate and refine the generated concepts. The objective was to reduce the number of alternatives while ensuring alignment with the defined requirement specification. The selection process was conducted in multiple stages, combining qualitative screening with more detailed comparative evaluation. Initially, an elimination matrix was used to, as described above, perform an early screening of the generated concepts, see figure 4.12. This step aimed to remove clearly unsuitable alternatives before a more detailed evaluation was conducted. The criteria used in the elimination matrix were derived from key requirements and included factors such as vertical adjustability, sensory feedback, valve operation, compact volume, and cost control. Concepts that did not meet these fundamental criteria were excluded from further development. This initial screening reduced the number of concepts and highlighted areas where additional information was required, particularly regarding technical feasibility. The remaining concepts were considered viable but required further development and refinement.

Elimination matrix						Chalmers	
Issued by: PEXA Project Team							
Created: 2026-02-23							
Solution	Elimination criteria						Decision criteria:
	1 The mouthpiece height shall be adjustable within a vertical range of 0-30 cm.	2 At least 90% of users shall correctly recognize and interpret the sensory feedback provided during operation.	3 The valve shall be operable with a single action and require a maximum force of 1.5 N	6 The total system volume shall be less than 0.125 m ³	11 Production Cost doesn't surpass current cost	Elimination criteria: (+) Yes (-) No (?) Search for info (!) Control productspec.	(+) Complete solution (-) Eliminate solution (?) Search for info (!) Control productspec.
						COMMENT	DECISION
Concept 1	!	+	+	+	+	It's unclear whether the long tube will work	?
Concept 2	!	+	+	+	+	It's unclear whether the long tube will work	?
Concept 3	+	+	+	+	+		+
Concept 4	+	+	+	+	+	Unsure about the sensory feedback from leaflet	+
Concept 5	+	+	+	-	-	Voice Control might be challenging and scissor lift	-
Concept 6	+	+	+	+	+		+
Concept 7	+	+	+	-	-	Requires more volume and additional weight/cost	-
Concept 8	!	+	+	+	+	It's unclear whether the long tube will work	?
Concept 9	!	+	+	+	+	It's unclear whether the long tube will work	?
Concept 10	+	+	+	-	-	Pneumatic solution is hard and expensive to implement	-
Concept 11	!	+	+	+	+	It's unclear whether the long tube will work	?
Concept 12	+	+	+	+	+		+
Concept 13	-	+	+	+	+	Mouthpiece not really height adjustable	-

Figure 4.12: Elimination Matrix

Following this early screening, a more detailed comparative evaluation was conducted using the Pugh matrix method. In the first iteration, the developed concepts were compared against the existing PEXA machine, see Figure 4.13), which served as a reference solution. Each concept was evaluated relative to the reference across a set of criteria derived from the requirement specification, using a qualitative scoring system. The results of this first iteration identified a subset of top-performing concepts that demonstrated clear improvements compared to the reference.

Chalmers <i>Issued by: PEXA Project group</i>	Pughmatrix Created: 2026-03-13 Modiafie						
	Current	THE DISHWASHER	BABA YAGA	LEADACOR	THE MINIMALIST	THE CLOSET	COMPACT
Criteria							
Requirements							
The mouthpiece height shall be adjustable within a vertical range of 0-30 cm.		+	+	+	+	+	+
At least 90% of users shall correctly recognize and interpret the sensory feedback provided during operation.		0	+	+	+	+	+
The valve shall be operable with a single action and require a maximum force of 1.5 N		+	+	+	+	0	+
The total system volume shall be less than 0.125 m³		0	-	+	+	+	+
Wishes							
Mobility: The system shall be movable by one person without mechanical lifting assistance		0	0	+	0	0	+
Production Cost doesn't surpass current cost		-	-	-	-	-	-
Weight: The total system weight shall not exceed 30 kg.		-	-	0	+	+	+
Warm-up Time: The system warm-up time shall not exceed 30 minutes.		+	0	+	+	+	+
Proper Installation Indication: The system shall provide a clear mouthpiece installation indicator that is correctly recognized by the operator.		+	+	+	+	+	+
Learnability: A first-time user shall be able to understand and complete the full operating procedure within 1-3 minutes, without external assistance		+	+	+	+	+	+
Σ+		6	5	8	8	8	8
Σ 0		4	2	1	1	2	0
Σ -		2	3	1	1	1	1
Net worth		4	2	7	7	7	7
Ranking		3	4	1	1	1	2
Further Development							

Figure 4.13: First Pugh Matrix

Based on this outcome, one of the leading concepts was selected as a new reference for a second iteration of the Pugh matrix, see figure 4.14. This allowed for a more refined comparison between the remaining concepts, highlighting their relative strengths and weaknesses in greater detail.

Chalmers	Pughmatrix	
Issued by: PEXA Project group	Created: 2026-03-13	
	Modified	
Criteria	LEADACOR	THE DISHWASHER BABAYAGA THE MINIMALIST CLOSET COMPACT
Requirements		
The mouthpiece height shall be adjustable within a vertical range of 0-30 cm.	0	0
At least 90% of users shall correctly recognize and interpret the sensory feedback provided during operation.	-	-
The valve shall be operable with a single action and require a maximum force of 1.5 N	0	0
The total system volume shall be less than 0.125 m³	-	0
Production Cost doesn't surpass current cost	+	+
Wishes		
Mobility: The system shall be movable by one person without mechanical lifting assistance (Service)	0	0
Weight: The total system weight shall not exceed 30 kg.	-	0
Warm-up Time: The system warm-up time shall not exceed 30 minutes.	0	0
Proper Installation Indication: The system shall provide a clear mouthpiece installation indicator that is correctly recognized by the operator.	0	0
Learnability: A first-time user shall be able to understand and complete the full operating procedure within 1-3 minutes, without external assistance	-	-
Σ+	1	1
Σ0	5	7
Σ-	4	2
Net worth	-3	-1
Ranking	3	1
Further Development	No	Yes

Figure 4.14: Second Pugh Matrix

Through this iterative evaluation process, lower-performing concepts were gradually eliminated, resulting in a smaller set of promising alternatives suitable for final assessment. To support a more objective and quantitative comparison, the remaining concepts were evaluated using a Kesselring matrix, see figure 4.15. In this method, evaluation criteria were assigned weights based on their relative importance, reflecting key aspects such as usability, technical performance, and feasibility. Each concept was then scored against these criteria, allowing for a weighted comparison of overall performance.

Chalmers		Kesselring Matrix											
Criteria		Ideal				Concepts							
		LEADACOR		THE MINIMALIST		CLOSET		COMPACT					
Issued by: PeXa Project team		Value	Grade 1-5	Result	Grade 1-5	Result	Grade 1-5	Result	Grade 1-5	Result	Grade 1-5	Result	
Ease of patient use		0,300	5,0	1,500	4,5	1,350	3,500	1,050	4,000	1,200	4,000	1,200	
Ease of operation		0,200	5,0	1,000	4,0	0,800	3,500	0,700	3,500	0,700	3,000	0,600	
Ease of serviceability		0,125	5,0	0,625	4,0	0,500	3,000	0,375	4,500	0,563	3,000	0,375	
Complexity of manufacturing		0,100	5,0	0,500	2,0	0,200	3,000	0,300	3,500	0,350	4,000	0,400	
Manufacturing cost		0,125	5,0	0,625	2,0	0,250	3,000	0,375	3,500	0,438	2,000	0,250	
Volume optimization		0,100	5,0	0,500	3,0	0,300	3,500	0,350	3,500	0,350	2,000	0,200	
Hygiene		0,050	5,0	0,250	4,0	0,200	4,000	0,200	4,000	0,200	3,000	0,150	
T (Totalt vikrat värde)		1,000	35,0	5,000	23,5	3,600	23,500	3,350	26,500	3,800	21,000	3,175	
T / T-ideal			1,0	1,000	0,7	0,720	0,670	0,670	0,760	0,760	0,635	0,635	
Medel					2,4		2,438		2,813		2,250		
Rank					2		3		1		4		

Figure 4.15: Kesselring Matrix

The results of the Kesselring matrix identified one concept as the most balanced solution, achieving the highest overall score across the weighted criteria. This concept demonstrated strong performance in both user-related aspects and technical feasibility, making it the most suitable candidate for further development. Overall, the multi-stage selection process enabled a systematic transition from a broad set of initial concepts to a single, well-justified design solution. By combining qualitative screening with iterative and quantitative evaluation methods, the process ensured that the final concept was both aligned with user needs and feasible from an engineering perspective. By applying this structured matrix evaluation process the generated concepts were systematically scored against the core requirements. The specific outcomes of this evaluation along with the final winning concept are presented in Chapter 5.

4.5 Selected Concept

As detailed in Section 4.4 of Chapter 4, the six alternative concepts were evaluated using Pugh and Kesselring matrices. By applying the specific weighting criteria established in the methodology, such as patient and operator usability, the final Kesselring evaluation identified the Closet concept described in Section 4.3.5 as the highest scoring and most viable design. Concepts such as the Leadacor and the Minimalist also scored closely and performed better in criteria such as ease of patient use.

The Closet concept was selected primarily due to its overall strong score, as well as its tall vertical structure and outward-opening doors, which provide improved ergonomic access for service personnel while also minimizing the external footprint. The evaluation also showed that several specific features from the other concepts were highly innovative and should not be discarded.

Therefore, the most suitable approach moving forward is to use the Closet concept as the foundational architecture while integrating the strongest elements from the other proposals to develop a final comprehensive redesign of the PExA device. By building on the Closet concept and integrating these additional features the final combined design is expected to meet the project requirements through a balanced approach to usability and technical performance. The compact external footprint fulfills the physical space limitations of standard health centers while the integrated digital guidance system improves the ease of operation for both the patient and the operator.

By replacing heavy sheet metals with injection molded recycled plastics and aluminum extrusions the product achieves both environmental sustainability targets [11] and the necessary reductions in cost and weight for clinical adoption [4]. Furthermore the retention of the patented inertial impactation core [1], [32] ensures that the device maintains the clinical accuracy required for reliable biomarker collection [27]. Overall the developed concept successfully bridges the gap between advanced respiratory research and accessible primary care screening.

4.6 Complementary Services and Value Proposition

To complete the product offer various complementary services have been developed to enhance user value and optimize the clinical workflow. The existing system already utilizes software to track live particle sampling rates and the particle count per individual breath [1]. By integrating this real-time data into a newly designed LCD screen featuring an optimized user interface both the patient and the operator receive immediate and clear feedback. This digital interface provides step-by-step breathing instructions such as when to inhale exhale or hold the breath as demonstrated in the prototype phase and can be synchronized with prompts for opening or closing the internal valve. This direct guidance ensures the complex breathing maneuver is performed correctly which generates cleaner and more reliable diagnostic data.

In addition to the physical interface a comprehensive software service integrates the device directly with electronic health records complying fully with the upcoming European Health Data Space regulations [20]. This integration enhances user value by streamlining the diagnostic workflow for healthcare personnel. Looking toward future scalability as the clinical user base expands PExA has the potential to integrate the sample analysis process internally. By centralizing this analysis the company can build a large reference database of respiratory biomarkers. This extensive data collection will not only help researchers discover new connections to various respiratory diseases but

also provide clinical healthcare professionals with highly accurate and verified diagnostic results [27].

To support the daily operation of the device a subscription model for disposable impactors and reservoirs will be implemented. This service ensures that clinics consistently maintain the necessary supplies without administrative delays. Furthermore to reduce environmental impact and resource consumption a dedicated recycling program for single-use plastic consumables will be introduced aligning with regional sustainable procurement strategies [11]. This initiative allows clinics to return used collection plates and other disposable components for safe and sustainable material recovery reducing costs while improving environmental sustainability.

4.7 Manufacturing Methods and Adaptability

The transition from a low volume research instrument to a widely used clinical device requires adapting manufacturing methods to enable scalable production. While the physical verification of new manufacturing processes falls outside the time constraints of this specific project phase, the conceptual redesigns were intentionally developed to support this future transition. The company currently relies on external contract manufacturers and engineering partners for device assembly and technical development [29].

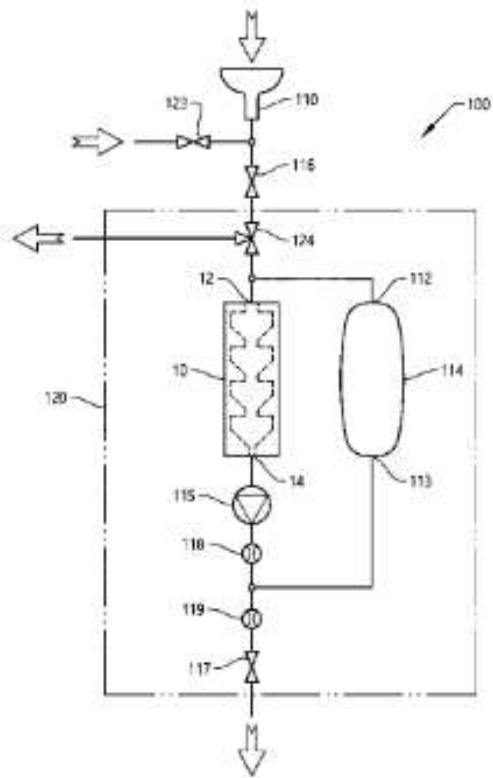
To produce the proposed concepts efficiently, the manufacturing strategy shifts from low volume assembly to standardized mass production. Based on the evaluations from the morphological matrix, the external casing of the new device concepts is designed to be manufactured using injection molding for the plastic components, replacing the heavier sheet metal of the previous iteration. The internal structure leverages standard aluminum extrusions to ensure stability while simplifying the building process. Standardized internal components such as vacuum pumps and optical particle counters will continue to be sourced from existing suppliers to maintain the core functionality.

Furthermore, to meet the requirements of contract manufacturers, the product design should intentionally minimise the number of custom fasteners and utilize standardized electronic connections to simplify assembly complexity [33]. During the ideation phase, alternative production methods such as Additive Manufacturing were considered for the external casing. However, this method was ultimately deemed too slow and inefficient for the projected annual volume required for a commercial clinical device, prioritizing injection molding as the superior manufacturing method [34].

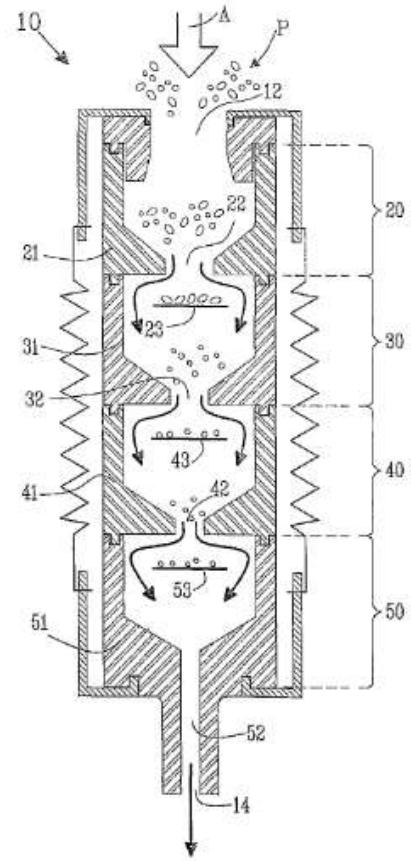
4.8 Patents Analysis

An analysis of relevant patents ensures that the redesign respects existing intellectual property while identifying new patentable features. For the PExA company, the patent protecting their non-invasive sampling technology is a key resource. It provides a distinct competitive advantage for future development and reduces the need for invasive diagnostic procedures such as biopsies [27].

The primary technology of the instrument is protected by several key patents identified through patent searches conducted in Espacenet. The main particle collection method from exhaled breath sampling, which utilizes an impactor, a vacuum pump, and a reservoir, is protected under patent US20100297635A1 [32]. Additionally, the methods for sorting and converting these collected particles into measurable data fall under patent US11534082B2 [35]. This specific patent explains how the PExA machine separates particles by size to ensure correct sampling to the substrate collection, which is fundamentally important for obtaining accurate biological information.



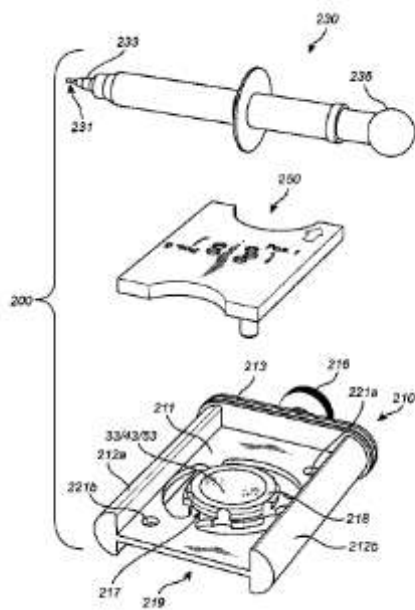
(a) Patent US20100297635A1 [32]



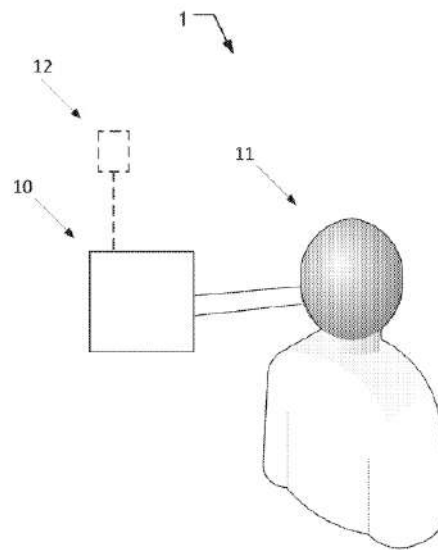
(b) Patent US11534082B2 [35]

Figure 4.16: Key patented mechanical components of the PExA device.

Furthermore, the sample preparation process is described in patent US12492972B2 [36]. This patent features a specialized device for holding and securing the collection plate alongside a punch cutting device integrated with a sample collector. Beyond the mechanical components, the application of this technology for detecting specific respiratory infections is covered by patent US20240423497A1 [37]. This patent details the capability of the device to recognize various respiratory diseases such as COPD, lung inflammation, and COVID 19. By analyzing the breath sample of a patient, specific particle variations and amounts can be detected to create a unique data profile for each individual. This patient data can then be compared against a larger reference database. A comprehensive big data collection supports the clear identification of whether a patient has a lung infection and specifies the exact type of infection present.



(a) Patent US12492972B2 [36]



(b) Patent US20240423497A1 [37]

Figure 4.17: Patented sample preparation device and diagnostic methodology.

These existing patents place specific restraints on the product redesign by requiring that the foundational sampling mechanisms must remain unchanged. Modifying this core technology falls outside the scope of the product development project, as it represents the defining feature of the PExA system and is therefore not the primary focus of the redesign. However, the new ideas generated from the six product concepts could present opportunities for new utility patents. For example, the newly designed automated guidance interface featuring an LCD screen and the redesigned compact internal component layout could be patentable if implemented. Additionally, if the long tube method proposed in several concepts proves successful, it presents another strong patent opportunity. This specific patent would cover the optimized thermal insulation of the tube, which prevents condensation by controlling the tube temperature to perfectly match the internal temperature of the machine.

4.9 Ethical Considerations

Introducing a new lung cancer screening device in primary healthcare centers raises several critical moral dilemmas and ethical considerations. A primary concern involves the potential for false positive results. False positives can cause significant patient anxiety and may lead to unnecessary invasive follow-up procedures, exposing patients to avoidable radiation. Therefore, ensuring high diagnostic accuracy and building a comprehensive reference database are essential to minimize these risks.

From one ethical perspective, offering early detection before visible structural changes occur can significantly improve patient outcomes by identifying disease at a stage where it is most treatable. Implementing this technology within a national screening program supports this objective. However, another perspective emphasizes the psychological burden placed on patients. If the initial PExA breath test indicates a potential risk, the patient must undergo a potentially stressful waiting period before receiving confirmatory structural imaging at a hospital, such as a LDCT scan.

An ethical statement guiding this product asserts that the device must be used strictly as a complementary risk assessment tool rather than a definitive diagnostic conclusion. The device cannot guarantee a lung cancer diagnosis with absolute certainty. Therefore raw test results should only be reviewed by medical professionals and never displayed directly to the patient on the device screen. This ensures that patients receive their results alongside comprehensive medical counselling from a doctor. Furthermore, data privacy remains an ethical imperative requiring

strict adherence to GDPR to protect sensitive biological information [19].

4.10 Technical Bottlenecks and Prototyping Focus

Despite the overall success of the conceptual redesign, several technical challenges and unknowns remain that require physical verification of the product concepts. The requirement to reduce the external physical footprint introduces a primary technical difficulty regarding the restricted internal volume. It is highly challenging to optimize this dense internal component packaging without the opportunity to physically disassemble and reassess the core internal parts, such as the vacuum pump and necessary electronics. Furthermore, reducing the overall physical volume creates internal thermal uncertainties. It is currently unknown whether a smaller housing requires additional silicone heaters to maintain stable temperatures, or alternatively, requires active ventilation and cooling. Because of project time restrictions and the practical difficulties associated with disassembling the existing machine, a complete physical redesign of the internal component structure won't be tested in the prototyping phase.

Similarly, the implementation of a custom printed circuit board (PCB) is an important theoretical consideration but remains out of scope for the physical testing. Integrating a custom board demands a comprehensive understanding of all internal and external interfaces, such as sensor signal processing, airflow valve control, power distribution, display control, particle counter interfacing, temperature monitoring, and actuator control. Due to mostly strict time limitations and the requirement for specialized electronic knowledge, this complex integration will not be prototyped.

Instead of treating the matrix selection as a final, linear decision, the project moved into a parallel development loop to address these bottlenecks. The highest scoring concept served as the primary baseline to detail individual subsystems, while specific mechanisms from the runner up concept were refined at the same time. This parallel workflow allowed the team to develop, test, and compare alternative sub solutions side by side. By iterating between both design paths, individual components could be evaluated on their own technical merits, making it possible to select and combine the best performing mechanisms before freezing the design for the prototyping phase.

Consequently, the physical prototyping phase detailed in chapter 5 focuses primarily on validating critical user experiences that cannot be fully resolved through digital modeling alone. For instance, the proposed height adjustment mechanisms introduce several challenges. The motorized lead screw provides precise control but increases manufacturing complexity, overall weight, and the risk of electronic failure. In comparison, the extended flexible tube approach represents an unverified solution with uncertainties regarding thermal loss and particle condensation, which could critically compromise the diagnostic sample.

To achieve validation within strict time constraints, the prototyping efforts prioritize the most important interactions for the patient, the operator, and the service personnel. For the patient, the primary focus is mouthpiece comfort and adjustability. Therefore, the prototyping phase will directly test, verify, and compare the long flexible tube method against the motorized lead screw mechanism. For the operator, the integration of an LCD screen will be evaluated to test the improved user interface, specifically focusing on simultaneous breathing instructions using custom designed visual prompts. Finally, to support the needs of service personnel, the outward opening hinged doors from the Closet concept will be tested to provide easier physical access to internal components.

A complete physical redesign of the internal component structure remains untouched in the prototyping phase. This decision stems from project time restrictions and the practical difficulties associated with disassembling the existing machine. However, previous interviews with service personnel confirmed that improving internal space organization and establishing structured maintenance and assembly instructions remain key requirements for the final theoretical design.

Chapter 5

Result from Prototyping and Testing

This chapter describes the implementation of the selected concept and the development of the final prototype. The prototype was constructed to evaluate how well the proposed design fulfilled the identified requirements. Through prototype testing and user evaluation, feedback was collected regarding usability, ergonomics, and functionality. The results provide an indication of the strengths and limitations of the proposed solution and form the basis for further development.

5.1 Prototyping

The prototyping phase was structured around isolating and testing specific sub-systems independently rather than building the entire assembly at once. This modular approach allowed for the verification of critical mechanisms without the interference of external variables.

The implementation followed a highly iterative process. When a specific prototype failed to meet the required functional criteria, it was analyzed and discarded. The project then progressed by building upon the concepts and physical components that had proven viable. This feedback loop ensured that individual parts worked reliably before moving toward the final physical integration.

5.1.1 Long tube

To evaluate the thermal performance and insulation capabilities of the long outside tube, an experiment was set up using an oven as the primary heat source. The Bill of Materials (BOM) for this test included foam noodles, silicone tubing, an oven, a digital timer and a two-probe thermometer.

Because of the tube's length, the oven door had to be kept open at a tilt to fit the assembly inside. To maintain the target internal environment of 37°C +/- 1.5°C the remaining openings around the door were sealed with isolating material. Temperature monitoring was handled by placing the first probe at the inlet of the tube and the second probe directly before the mouthpiece.

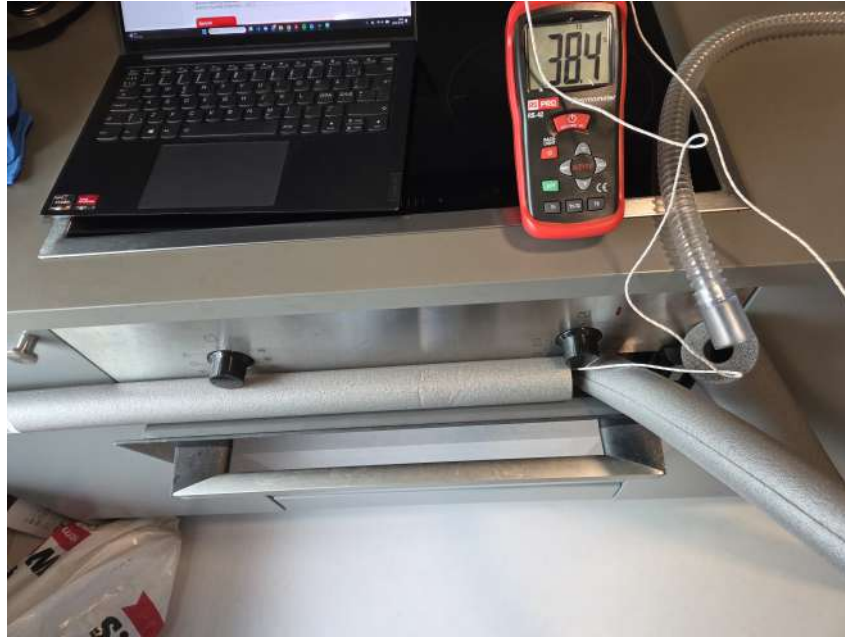


Figure 5.1: The Compact concept

The testing protocol evaluated three different lengths 90 cm, 60 cm, and 30 cm, with each trial running for a duration of 30 minutes. To test insulation efficiency, measurements were taken using both 9 mm and 20 mm foam thicknesses. Prior to each test run, the ambient room temperature was recorded to calculate the exact temperature increase across the setup.

These results demonstrate that a standard long tube design does not provide sufficient thermal performance for the system's operational requirements. In the subsequent discussion chapter, the limitations of this testing methodology will be analyzed. Specifically, using an oven instead of the actual machine introduced significant environmental variables; heat leaked through the open sides of the oven door, warming the exterior walls of the tubing rather than heating the air strictly from the inlet. Additionally, maintaining a perfectly steady ambient baseline proved difficult due to small air drafts from an open window, which was required to regulate the room's temperature.

5.1.2 Lead Screw Mechanism

In parallel of developing the long tube concept members of the team designed in CAD the leadscrew assembly and then proceed to assemble a physical prototype. It consists of multiple parts that work together in order to translate the rotational motion of the stepper motor into a linear motion which in turn moves the position of the mouthpiece vertically.



Figure 5.2: Render of Lead Screw Assembly

5.1.3 Part Assembly

To understand the importance of each part and the reasons that they were selected, it is important to explain the assembly of this mechanism. The widely available Nema 17 Stepper motor with a maximum torque of 510 Nm was used, which provides smooth rotational movement with precise positioning and fixed angular steps of 1.8 degrees per step, giving 200 steps/revolution. The stepper motor provides high torque in low speeds which is ideal in the case as precision is favorable when compared to speed. When stationary and energized, it holds its position with significant torque as this is valuable in a lead screw assembly to resist back-driving when vertical load forces are applied from the patient, without the need of additional braking mechanisms. The stepper motor requires a dedicated driver module to function, as the control signals generated by the microcontroller are insufficient to directly energise the motor coils.

For this assembly, the Trinamic TMC2208 stepper driver was selected, as it supports a supply voltage range of 5 to 36 V and is capable of delivering up to 2 A of peak current per output which is well suited for the selected stepper motor which has a maximum current of 1.2 A. A key feature of the TMC2208 is its inaudible motor operation both during motion and at standstill making it beneficial in environments such as Health Centers.

A shaft coupler in the assembly physically connects the motor shaft to the leadscrew, transmitting rotational motion from one to the other. It can also accommodate small degrees of angular, parallel, and axial misalignment between the two shafts, preventing the transmission of bending loads into the motor bearings or the leadscrew, which would otherwise cause premature wear or binding.



Figure 5.3: Lead Screw Physical prototype - 1st Iteration

A hardened steel lead screw with a 2 mm pitch drives a bronze nut securely mounted on a custom 3D-printed base. This base is designed to accommodate the flexible tube which connects to the inner valve and the mouthpiece on the exterior of the machine. Two hardened steel guide rods constrain the rotational motion of the bronze nut, translating the stepper motor's rotation into precise linear movement. The base plate is fitted with two linear bearings to ensure smooth travel along the rods and minimize friction between components. To avoid placing unnecessary radial loads on the stepper motor shaft, the motor is mounted at the top of the assembly with its shaft facing downward, preventing any efficiency losses. The motor is secured using a custom 3D-printed PETG bracket fastened with M3 screws. Two additional 3D-printed bases, each fitted with a central bearing, hold the lead screw and guide rods in fixed positions while allowing smooth, low-friction rotation of the lead screw. The entire assembly is mounted onto two aluminum plates, which provide structural support and rigidity to the system.



Figure 5.4: Lead Screw Render (CAD 1)



Figure 5.5: Lead Screw Render (CAD 2)

5.1.4 Electronics

The control system of the leadscrew assembly is centred around an Arduino Uno R3 microcontroller, which was selected for its ease of programmability via the Arduino IDE, a development environment based on the C++ programming language. The microcontroller coordinates the behaviour of all input and output components within the system, including the stepper motor driver, limit switches, and push buttons.

Directional control of the mouthpiece carriage is achieved through two momentary push buttons, one assigned to each direction of travel. Momentary push buttons are characterised by their ability to close the circuit only whilst being actively depressed, returning to an open state upon release. This behaviour ensures that motor motion is continuous only for the duration of user input, providing intuitive and responsive manual control of the carriage position.

To prevent the carriage from exceeding the physical boundaries of the assembly, two limit switches are mounted at the uppermost and lowermost positions of the travel range. Upon contact with the carriage, each switch sends a signal to the Arduino, which immediately commands the stepper motor to cease rotation regardless of whether a directional button is being pressed. This hardware-level safeguard protects the mechanical components from damage due to over-travel.

Since the stepper motor operates at 24 V, a voltage that exceeds the 5 V logic supply provided by the Arduino, an external 24 V power supply is connected directly to the TMC2208 driver module, which in turn regulates and delivers the appropriate current to the motor coils. The digital input and output pins of the Arduino are used to interface with both the push buttons and limit switches, while the motor's four-wire bipolar winding configuration connects directly to the driver module's output terminals.

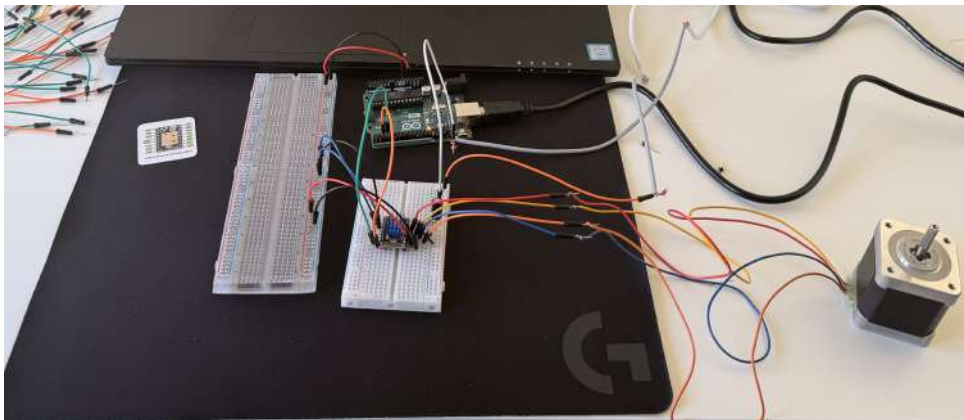


Figure 5.6: Assembly Of Components

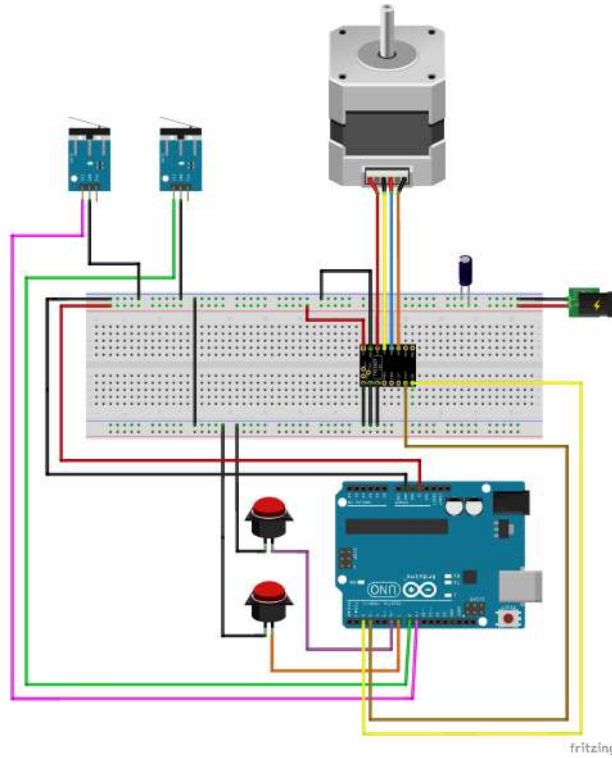


Figure 5.7: Schematics of Electrical Components

5.1.5 Structure of Prototype

Internal Structure

The structural frame of the prototype on which the lead screw assembly was also mounted is constructed from 30×30 mm T-slot aluminium extrusion profiles. Aluminium extrusions were selected for this application due to several properties that make them particularly well suited to medical device development. Their high strength-to-weight ratio provides the structural rigidity necessary to support the leadscrew assembly and guide rods without adding unnecessary mass.

The T-slot profile system enables a highly modular and flexible construction approach, allowing components to be securely fastened and repositioned without the need for welding or specialised tooling. This is a significant advantage during the iterative design and prototyping phases of development. Furthermore, the corrosion resistance of aluminium, combined with its smooth anodised surface, ensures the frame is both durable and easy to clean and disinfect, meeting the hygiene requirements inherent to medical device applications. The relative cost-effectiveness and machinability of aluminium extrusions also make them a practical choice for small-batch and prototype medical equipment compared to alternative materials such as stainless steel.



Figure 5.8: Frame Construction

As the assembly was developed to evaluate a predefined set of functional areas rather than to replicate the full internal layout of the existing PExA machine, the overall dimensions of the structural frame were adapted accordingly to suit the specific requirements of the prototype. Frame extrusions are arranged in a combination of vertical and horizontal members, with supporting beams connected at 45-degree angles to provide the necessary rigidity and stability. Since standard off-the-shelf connectors were insufficient to accommodate the specific angular junction, custom connectors were designed and subsequently manufactured using additive manufacturing, with PETG selected as the material due to its mechanical strength and suitability for functional prototype components.

Exterior Panels

Given that the assembly was still in the prototype phase, additive manufacturing was selected as the fabrication method for the exterior housing. The panels were designed with the intent that the final production version would be manufactured via injection moulding using ABS. However, for the purposes of this prototype, PETG was selected as the printing material due to its rigidity, dimensional stability, and relative ease of processing.

A significant constraint encountered during the manufacturing phase was the limited build volume of the available 3D printers, which imposed a maximum print bed size of $250 \times 250 \times 250$ mm which is considerably smaller than the overall dimensions of the housing. To address this, the exterior housing was subdivided into 15 individual panels, which were subsequently joined using M3 nuts and bolts. The panel geometry was adapted to incorporate recessed features to accommodate the nuts, enabling secure fastening between adjacent pieces. Additionally, custom attachment brackets and T-slot rollers were designed and manufactured separately to secure the panels to the aluminium extrusion frame. In total, the printing process consumed approximately 3 kg of PETG filament and required over four days of continuous print time.

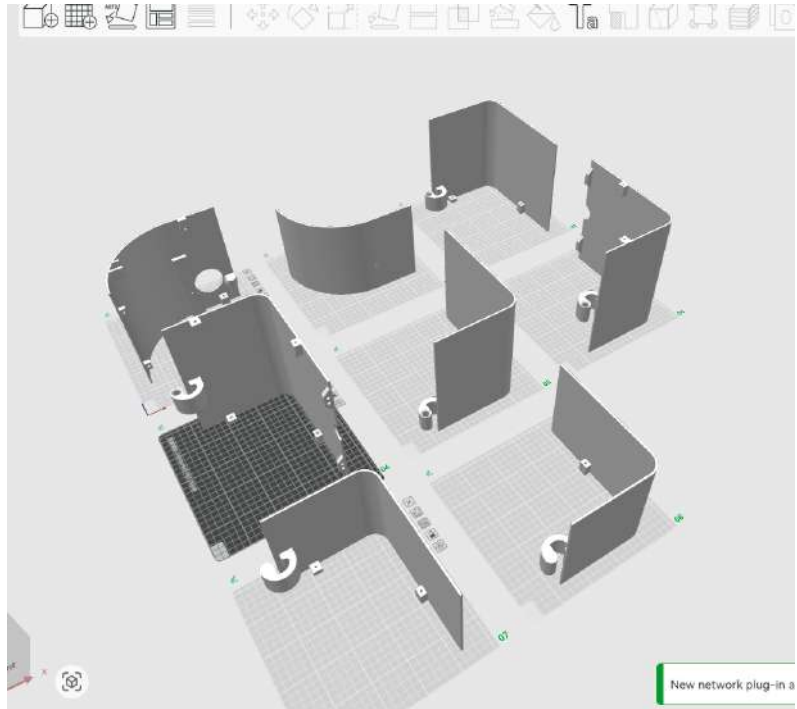


Figure 5.9: Panels in Slicer Program

Particular attention was given to dimensional tolerances throughout the design process, as the assembly of multiple printed components introduced the risk of cumulative deviations arising from material shrinkage and print misalignment. While these effects were largely mitigated through careful design, minor surface gaps between certain panels were observed as a result of localised warping during printing.

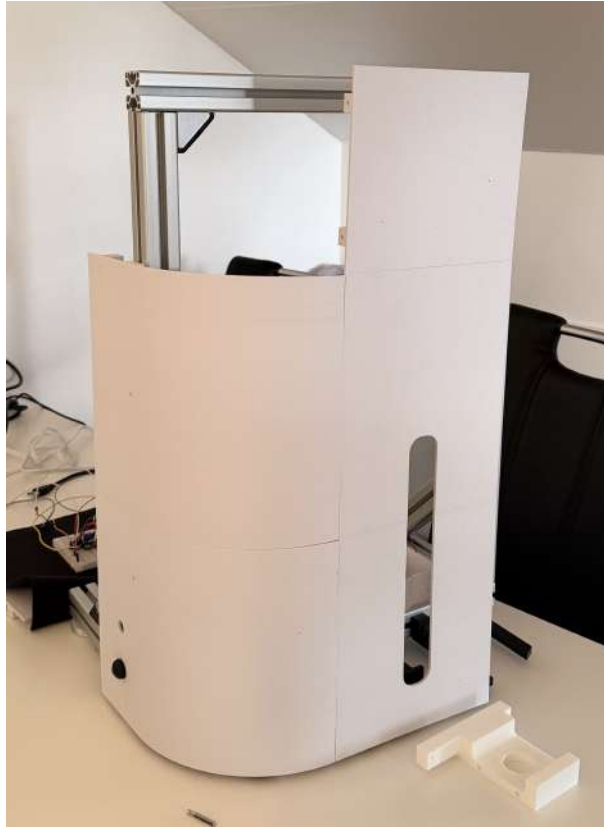


Figure 5.10: Exterior Panel Assembly

Hinged Doors

An important functional requirement of the prototype was the implementation of hinged back doors, designed to allow rear access to the internal components of the machine for the service personnel. The doors were conceived to rotate outward, with the hinge mechanism fully concealed within the frame so that the exterior surface of the panels remained flush, resulting in a cleaner aesthetic and a surface that is easier to disinfect. Each door consists of three panels joined together to act as a single rigid door leaf.

For the final production version, off-the-shelf stainless steel hinges would be specified to meet the durability and hygiene requirements of a clinical environment. However, given the constraints of the prototype phase, a custom hinge mechanism was instead realised through the geometry of the 3D printed panels themselves. Specific panels were designed with an integrated cylindrical protrusion that served as a pin axle, around which the corresponding door panels could be mounted and rotate freely. A minor clearance gap was maintained between adjacent panels to prevent surface contact and collision during opening and closing. This approach allowed the hinge functionality to be achieved without any additional hardware, whilst keeping the exterior profile of the assembly clean and uninterrupted.



Figure 5.11: Hinge Door Mechanism

User Interface

The user interface of the prototype was initially designed around a capacitive touchscreen display intended to connect directly to the Arduino Uno R3 microcontroller, which would have allowed for a customisable interface developed using the screen manufacturer's proprietary software. However, a software incompatibility encountered during the development phase prevented successful implementation of the display within the system.

This setback also prompted a reconsideration of the screen's placement on the machine. In the original design, the display was to be mounted on the upper front face panel of the frame. However, it was determined that positioning the screen on the mouthpiece carriage base would provide a more ergonomic viewing angle for the patient during the test procedure. Since the phone holder was mounted directly onto the carriage base alongside the mouthpiece, the screen travels with the carriage during operation, ensuring it remains consistently oriented toward the patient throughout the procedure.

To create the interface for prototype testing purposes, a smartphone was used in place of the dedicated display, taking advantage of its high-resolution capacitive touchscreen. An interactive prototype of the user interface was developed in Figma, utilising its animation and interaction capabilities to produce a fully navigable simulation of the intended interface.

The procedure guides the patient through six introductory steps outlining the correct breathing method, after which the patient selects between two valve control modes. The first is a manual mode, in which the patient operates a side-mounted hand lever to open and close the valve, allowing them to dictate the duration of exhalation into the impactor according to their individual breathing capacity. The second is a timed mode, in which a fixed time interval is pre-set for each breathing cycle, automating the valve operation for consistent particle collection. Following each completed breathing cycle, a progress indicator is displayed, informing the patient of how much of the overall procedure has been completed, and the cycle repeats until the required quantity of particles has been collected.

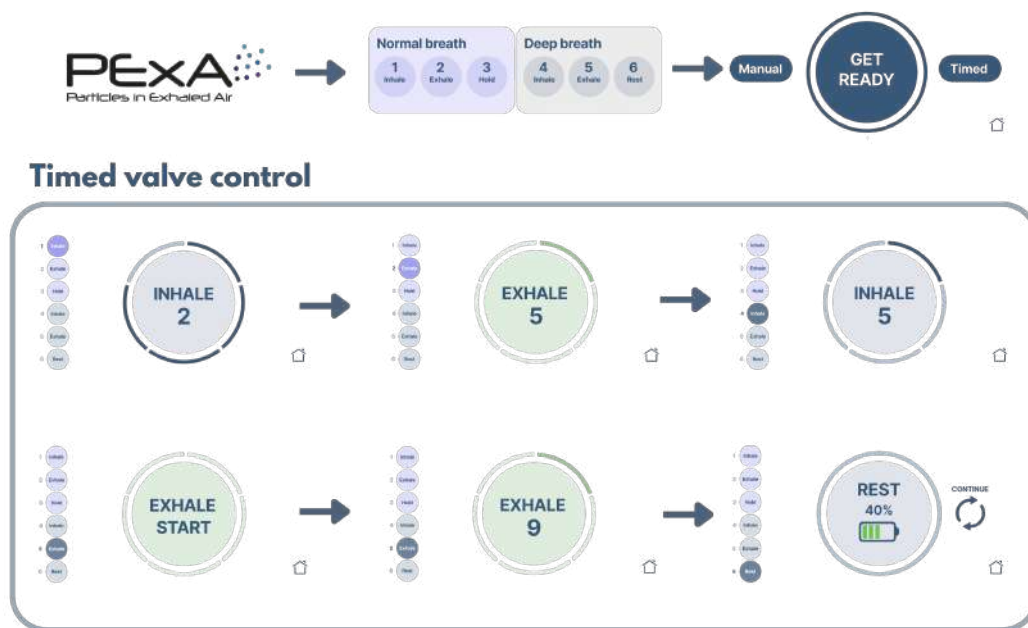


Figure 5.12: Timed Valve Control UI

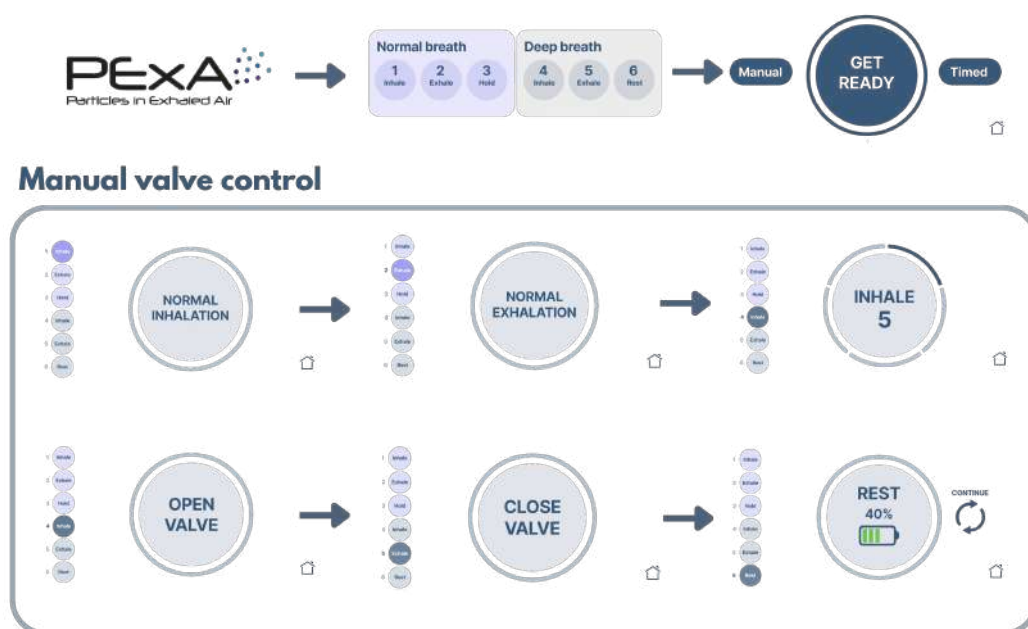


Figure 5.13: Manual Valve Control UI



Figure 5.14: Manual Valve Control UI

5.2 User Evaluation

To evaluate the system's usability and ergonomics, the prototype was put through user testing during the public exhibition. This setup allowed various individuals with different physical profiles to interact with the product, providing a broad overview of how the design adapted to different users.

The evaluation combined direct observation of user interaction with qualitative feedback. Observing the participants made it possible to identify how intuitive the product was and see if the physical interface accommodated different users naturally. Furthermore, the participants were interviewed afterward to gather their direct impressions and critiques. The data collected during this phase highlighted the practical strengths of the design and pinpointed specific areas needing final adjustment.

5.2.1 Prototype Feedback

The feedback pertaining to the leadscrew assembly and mouthpiece height adjustability was largely positive. Attendees were able to operate the two side-mounted push buttons with ease, intuitively adjusting the mouthpiece to their preferred height. The motion was noted to be silent, consistent with the expected behaviour of the TMC2208 driven stepper motor. However, several attendees expressed a preference for a faster traversal speed. This was subsequently addressed by modifying the relevant parameters in the control software. The angular orientation of the mouthpiece was well received, with attendees noting that the slight forward-leaning posture is encouraged, consistent with the clinically defined "tripod-orthopneic" [38] position.



Figure 5.15: Final Prototype

One notable limitation identified during the exhibition concerned the lower boundary of the mouthpiece travel range. Whilst the overall range of vertical displacement was considered sufficient, it was observed that the lowest reachable position of the carriage would need to be raised in order to accommodate the tube connecting the mouthpiece to the HEPA filtration system, which filters the air inhaled through the mouthpiece. This spatial requirement had not been fully accounted for in the current design iteration, and it is recommended that future development addresses this constraint more thoroughly during the layout and dimensioning phases. Overall, the leadscrew assembly performed as intended throughout the exhibition, and the feedback received validated the core functionality of the height adjustment mechanism.

Feedback regarding the user interface was generally positive, with attendees finding the overall experience comfortable and intuitive. The screen distance of 20 cm from the eyes was a deliberate ergonomic design decision, corresponding to the minimum focal length at which the eyes can comfortably focus on a screen without strain. To accommodate users who may prefer a greater viewing distance, alternative mouthpiece tubes of varying lengths were printed and made available for testing. However, it was observed through user testing that the original 20 cm distance was consistently preferred among attendees, validating the initial design choice.

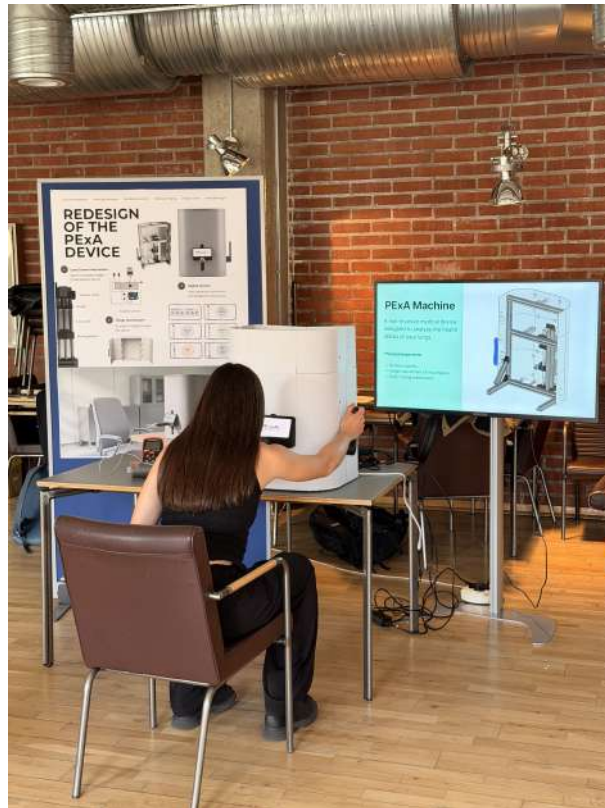


Figure 5.16: Attendee using the Prototype

The dual valve control modes were well received, with the mixed demographic of attendees highlighting the relevance of offering both options. Younger attendees with greater lung capacity tended to favour the manual control mode, as it afforded them greater flexibility over the duration of each exhalation cycle. Older attendees, whose lung capacity may be more limited, found the timed mode more manageable, as it provided a structured and predictable breathing cycle. The simplicity of the Figma interface and the clarity of the on-screen instructions were considered adequate for guiding users through the procedure. It was noted, however, that the addition of audible feedback cues would further enhance the user experience, and that for certain user groups, supplementary verbal guidance was beneficial in clarifying the procedure and resolving any ambiguities.



Figure 5.17: Prototype

Regarding the hinged back doors, a comprehensive evaluation of their functional effectiveness in providing access to the internal components was not possible at this stage, as the internal components had not yet been installed within the prototype. Nevertheless, when the concept was explained to attendees, the hinged door mechanism was broadly preferred over the alternative of fully removing the exterior panels to gain access, suggesting that the approach is well aligned with user expectations and should be carried forward into future development iterations.

Chapter 6

Commercial Assessment of the developed product

This chapter provides a comprehensive commercial assessment of the PExA system. Currently, PExA is positioned primarily within the research market, characterized by broad protein analysis, complex manual workflows, and high experimental costs. The strategic advantage of this positioning is the relative regulatory flexibility, as it is not yet subject to the strict certifications required for clinical diagnostic devices, and the customer base is less price-sensitive and more focused on experimental versatility.

For the next phase of growth, PExA aims to transition from a specialized research tool to a scalable commercial diagnostic solution. The ultimate objective is to integrate PExA into national screening programs, thereby capturing a significant share of the diagnostic market. However, this transition poses a substantial cost challenge: current broad biomarker analysis costs approximately \$250 per sample. To be commercially viable, the target cost could be reduced to dozens of dollars by focusing only on a few specific biomarkers.

Consequently, the primary hurdles for market entry are identifying specific, high-value biomarkers and optimizing the sampling workflow. Furthermore, the hardware cost must be addressed since the current unit price of \$75,000 remains a significant barrier for local health centers.

In the following sections, the context operate under the assumption that PExA will successfully isolate specific and clinically relevant biomarkers and transition toward full-scale commercialization. Various business models designed to enhance price-to-value perception for customers were expored. Finally, the chapter concludes with a strategic roadmap for market adoption and a detailed risk analysis.

6.1 Market Prerequisites

This section outlines the situations and barriers when PExA tends to enter the commercial medical market in Sweden. It considers factors such as laws, market demands, and market competitiveness. Understanding these prerequisites is essential for positioning the product successfully within its intended market.

6.1.1 Regulatory Compliance

To achieve market entry, PExA is legally required to conform to the EU Medical Device Regulation (MDR 2017/745), a framework enforced locally by the Swedish Medical Products Agency (Läkemedelsverket) [39]. Aligned with this regulatory compliance, the project strategically prioritizes the Swedish market as its initial commercial launchpad. A critical task in this phase is not only achieving CE (Conformité Européenne) Marking compliance, but also fulfilling the specific localization requirements mandated. This includes ensuring that all user

interfaces (UI) and instructions for use (IFU) are fully adapted into Swedish, which is paramount for maintaining clinical safety and ensuring local regulatory adherence.

This regulatory strategy inherently supports significant future scalability. Due to the harmonized legal framework of the EU Single Market, once PExA secures CE Marking and satisfies Swedish market access requirements, the product gains a high degree of regulatory leverage across the entire European Economic Area (EEA). Consequently, this strategic approach effectively lowers barriers for further expansion, and future entries into other EU member states will be streamlined, avoiding redundant regulatory processes, optimizing time-to-market, and reducing the operational costs associated with compliance. This foundation allows PExA to efficiently scale its presence throughout Europe [40].

The market in the United States is strictly governed by risk-based classification and the establishment of product equivalence. Given its non-invasive mechanism for collecting biomarkers from small airways, PExA is expected to be categorized as a Class II (moderate risk) medical device. To achieve commercial clearance, the project faces two distinct regulatory pathways depending on the competitive landscape database [41]:

- **510(k) Premarket Notification:** If a legally marketed predicate device with similar technological characteristics and intended use is identified, PExA must demonstrate *Substantial Equivalence* (SE). This pathway focuses on demonstrating that the device is as safe and effective as the predicate, thereby significantly optimizing time-to-market.
- **De Novo Classification Process:** In the absence of a suitable predicate device due to PExA's novel aerosol collection technology, the project must utilize the De Novo pathway. This petition allows for a risk-based evaluation to classify a novel device into Class I or Class II, which necessitates more extensive clinical validation data and longer regulatory review timelines.

Furthermore, as a foreign manufacturer established in Sweden, legal entry into the US market requires mandatory compliance with 21 CFR Part 807. This encompasses annual FDA Establishment Registration (incurring a recurring fixed regulatory expenditure) alongside formal Device Listing. Additionally, the project must designate a permanent US Agent to act as the legal communication between the company and the FDA, establishing a foundational administrative overhead for transatlantic commercialization [42].

6.1.2 Competitor Analysis

The current pulmonary diagnostic landscape includes tools such as FeNO analyzers [43], respiratory oscillometry [44], and spirometers [45], which primarily assess airway inflammation and lung capacity. While valuable for functional assessment, these methods cannot provide structural imaging of the lungs. Conversely, Low-Dose Computed Tomography (LDCT) serves as the clinical gold standard for structural imaging and is currently the primary method for early lung cancer detection. [46].

To evaluate the relative positioning of PExA 2.1, a benchmark comparison with LDCT was conducted (see table 6.1). LDCT was selected as the primary reference due to its prevalence in lung screening, despite the fundamental difference in detection mechanisms compared to PExA 2.1.

While LDCT is an established tool for structural imaging, it may struggle to identify subtle physiological changes in the earliest stages of lung cancer. In contrast, PExA 2.1 offers a non-invasive, radiation-free method to analyze biomarkers collected from the small airways, potentially identifying early-stage changes that precede structural manifestation. Furthermore, PExA 2.1 offers a more cost-effective screening approach compared to LDCT. From the interview with PExA, their goal is to determine the analysis price for each biomarker once a specific protein is confirmed to be relevant to lung cancer. Consequently, PExA 2.1 is positioned as a complementary technology that addresses the limitations of LDCT by acting as a triage tool, rather than as a standalone diagnostic device

Table 6.1: Comparison between PExA 2.1 and LDCT product, GE Revolution used as a representative example of a state-of-the-art LDCT system to illustrate the characteristics of imaging-based lung cancer screening [47].

Category	PExA 2.1	GE Revolution(LDCT)
Detection Type	Biochemical markers (particles)	Structural imaging
Object Detected	Molecular / protein changes from small-airway	Visible structural nodule
Radiation	None	Yes (low dose)
Invasiveness	Completely non-invasive	Minimally invasive (radiation)
Clinical Maturity	Emerging	Established, guideline-recommended
Limitations	Not a diagnostic cancer tool yet	Misses biochemical early-stage changes; false positives
Price	\$10 (estimated)	\$1,200

6.1.3 Demand Estimation

To estimate the total demand for PExA in Sweden, the data is that approximately 20–22% of the Swedish population falls within the 55–74 age range [48], while adult smoking prevalence is estimated between 6–10% [49]. By focusing on the intersection of these groups, those aged 55–74 with a significant smoking history, the result identifies a high-risk group of approximately 290,000 (10 million total population x aged from 55 to 74 x smoking group) individuals as the primary target for initial clinical triage. An examination shows that the total demand through two distinct adoption scenarios, as market penetration rates vary significantly depending on how the device is integrated into clinical practice.

In the first scenario, PExA acts as a high-risk triage tool outside of a national program. In this model, penetration is "bottom-up" and driven by individual clinical adoption, resulting in a gradual market uptake as providers recognize the device's value in reducing unnecessary imaging. Due to the decentralized procurement structure across Sweden's 21 autonomous Regions, market uptake is expected to be gradual. Assuming a conservative penetration rate of 10% during this initial growth phase, the annual demand is estimated at 29,000 tests (based on a 290,000 target population $\times$ 10% penetration).

In the second scenario, PExA is integrated into a national lung screening program. This "top-down" approach significantly expands the target group to the broader demographic of 55–74-year-olds, encompassing approximately 2,000,000 individuals. If PExA is adopted as a standard component of this national protocol, market penetration would be significantly higher, estimated at 80%, which is the case for breast cancer screening[50], since the device transitions from a discretionary tool to a required element of standardized care. This yields a total annual demand of 1,600,000 tests (2,000,000 target population $\times$ 80% penetration).

6.2 Financial Viability

This section presents a cost analysis of PExA 2.1, manufacturing, maintenance, and compliance regulations. The aim is to evaluate whether the product can be produced and sold at a competitive and sustainable price point.

6.2.1 Cost Breakdown

Because PExA 2.1 is a medical device, deeper research was done on compliance regulations. Based on the Swedish Medical Products Agency (Läkemedelsverket) [39], to maintain a lawful market presence in Sweden, the project must account for three core annual regulatory fees.

- **Economic Actor Role Fee (30,000 SEK/year):** A mandatory requirement under the EU Medical Device

Regulation (MDR 2017/745) for manufacturers to maintain their registration and traceability within the EUDAMED database.

- **Company Registration Fee (2,150 SEK/year):** A national administrative fee required to maintain the company's legal standing within the Swedish medical device registry.
- **Device Maintenance Fee (1,000 – 5,000+ SEK/year):** A scalable annual fee that fluctuates based on the total number of registered devices. For a portfolio of up to 10 devices, the fee is 1,000 SEK, increasing as the product line expands.

Collectively, these annual operational expenditures ensure that PExA remains compliant with both Swedish national legislation and the broader EU MDR framework.

Beyond the mandatory regulatory fees and compliance costs previously outlined, the project's financial viability is heavily dictated by the direct costs associated with the product itself, as well as the intensive capital required for its ongoing development and manufacturing. Based on primary data from PExA, the core financial structure is defined as follows:

- **Unit Manufacturing Cost(50,000 USD/unit):** The direct cost for producing and procuring a single PExA unit. This represents the fundamental capital expenditure required per unit before any market markup is applied.
- **Current R&D(100,000 USD/month):** While unit costs are fixed, the broader project is currently in a high-intensity Research and Development (R&D) phase. The monthly operational expenditure is currently 100,000 USD/month. This budget covers ongoing technical refinement and the extensive clinical validation required to bridge the gap toward a full commercial launch.

6.3 Profit Analysis

This section explores two business models, a servitization model and a one-time purchase model, over a seven-year horizon, which includes a one-year development period followed by six years of market commercialization. To simplify development-phase variables, it was assumed that specific biomarkers have already been identified within the commercial modules, removing the unpredictability of early-stage biomarker discovery.

To assess the financial potential of PExA 2.1, an NPV analysis was introduced. Since this market depends heavily on volatile factors like screening population and hardware sales volume, the NPV, illustrated as equation 6.1, is evaluated across worst, base, and best-case scenarios to capture the full spectrum of outcomes. These scenarios are built upon the projected revenues of each selling model, offset by the related costs of PExA 2.1. In order to completely simulate the commercial results, some factors are based on the assumptions.

$$NPV = \sum_{\text{periods}} \frac{\text{period cash flow}}{(1 + \text{discount rate})^{\text{period}}} \quad (6.1)$$

6.3.1 Business Model: Servitization

In the servitization model, PExA will charge a one-time fee to health centers and a screening fee each time a patient undergoes screening. This commercial model does not impose a heavy financial burden on health centers looking to adopt the machine; however, PExA's profit is strongly influenced by the screening population. Below, the details of each variable are discussed, and the NPV values for pessimistic, base, and optimistic scenarios are presented at the end.

- **Screening population:** Based on the demographic assessment conducted in the previous section, the total high-risk target population (N) is established at 290,000 individuals. To estimate the annual test volume

(Q), the demand forecasting formula $Q=(N \times A \times P)/f$, where f represents the screening frequency was applied. In alignment with Swedish clinical standards for national screening programs, such as biennial breast cancer screening, the screening frequency was set to once every two years ($f=2$).

Regarding the reachable and aware fraction (A), a conservative range of 10% to 20% was projected, a constraint directly attributed to the procurement decentralization across Sweden's 21 autonomous healthcare Regions. The adoption probability (P) is anchored in primary survey data, shown in Table 6.1, evaluating public interest in alternative screening methods. Empirical results indicate that 75% of participants expressed an immediate willingness to undergo non-invasive, radiation-free screening, with an additional 22.9% remaining open to the procedure, justifying an adoption probability range of 75% to 97.9%.

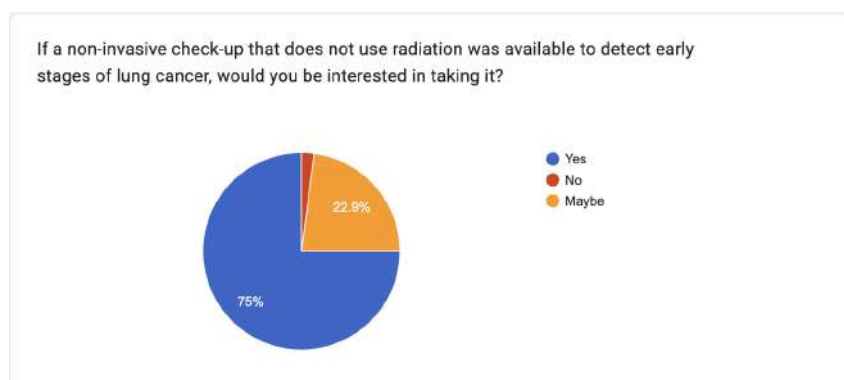


Figure 6.1: The Interest in Taking a Non-invasive Screening of Lung Cancer

Consequently, evaluating these variables under the biennial screening cycle yields three projected annual test volumes: 21,750 (conservative), 36,975 (base-case), and 13,775 (optimistic).

- **Rental adoption:** Based on the projected Reachable and Aware Fraction, the rental adoption rate is estimated to be between 10% and 20% of all primary health centers in Sweden. Assuming a total base of approximately 1,200 healthcare facilities nationwide, this translates to an initial target market of 120 to 240 active rental units.
- **Test price:** The clinical service fee per test is defined based on empirical willingness-to-pay data obtained from the public survey, shown in Table 6.2. The survey results indicate that 43% of respondents are willing to pay between \$1 USD and \$100 USD for the non-invasive screening, while an additional 36.6% are responsive to a higher price range between \$101 USD and \$200 USD. Anchored upon these consumer baseline preferences, the test price is projected across three commercial growth scenarios: \$50 USD for the pessimistic case, \$100 USD for the base case, and \$150 USD for the optimistic case.

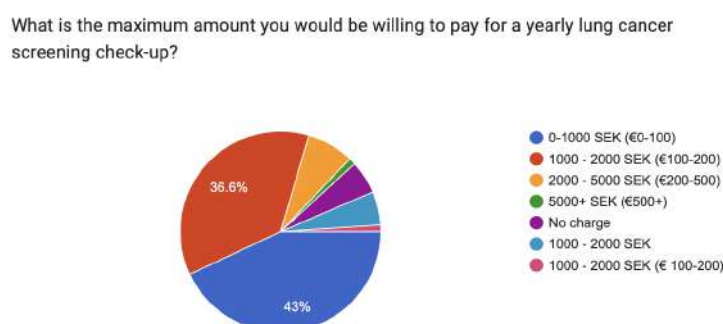


Figure 6.2: The Willingness of paying for an Invasive Screening of Lung Cancer

- **Maintenance cost:** It is assumed that the maintenance fee currently is \$300 USD every two years for one machine. For the three scenarios, the cost is established at \$150, \$300, and \$450 for one unit, respectively.
- **Product cost:** According to primary interview insights provided by PEXA, the diagnostic device currently commands a 50% mark-up at a retail price of \$75,000 USD, which establishes the baseline Cost at \$50,000

USD per unit. Recognizing that production expenses will likely be influenced by external market forces and supply chain dynamics upon commercial launch, the financial model implements a variance boundary to stress-test these projections. Under the optimistic (best-case) scenario, manufacturing optimizations are expected to lower the unit cost to \$45,000 USD, whereas macroeconomic pressures and potential material shortages under the pessimistic (worst-case) scenario are projected to elevate the unit expenditure to \$55,000 USD.

- **Development cost:** During the interview with PExA, they shared that they roughly spent \$0.1 million USD on development every month. It is assumed that after targeting specific bio marks, it takes three to seven years to bring a device from concept to approval [51], several development years were condensed into one year in the business model, so the cost of three different situations should be \$3.6, \$6, and \$8.4 million USD.
- **Marketing and support:** It was assumed that the marketing and support cost should be from \$0.3 million USD to \$0.5 million USD.
- **Initial access fee:** For public health centers, PExA will be more acceptable if the fee can be \$10,000 USD. Thus, the fees can be defined as \$5000, \$10000, and \$15000 USD.
- **Equipment and tooling:** PExA 2.1 utilizes a procurement strategy based on off-the-shelf components to minimize initial tooling requirements and simplify the supply chain. This strategy significantly reduces the need for custom tooling and development. By leveraging standardized parts, the project achieves a streamlined manufacturing setup, though direct procurement costs are integrated into the unit manufacturing cost analysis rather than being treated as negligible.

Combining all of the variables above, the net present values can be realized. From Table 6.3, the incomes eventually can't cover the cost, resulting in negative NPVs in the worst and basic scenarios and only making \$3.8M USD profits in the best. This financial deficit is primarily driven by the massive upfront development cost and product cost relative to the limited annual test volume and conservative test pricing.

Servitization			
Variables:	Worse	Base	Best
Screening Population (Person)	21750	36975	56782
Rental Adoption(Unit)	120	180	240
Test Price (USD/patient)	50	100	150
Maintainance Cost(USD/unit)	450	300	150
Discount Rate	0.09	0.07	0.05
Product Cost(M USD):	0.056	0.0500	0.046
Development Cost (M USD):	8.4	6	3.6
Marketing and Support (M USD):	0.5	0.4	0.3
Initial Access Fee (M USD):	0.005	0.01	0.015
Equipment and Tooling:	0	0	0
Net Present Value(Million USD)	-\$19.4	-\$11.4	\$3.8

Figure 6.3: Sensitivity Analysis and Financial Projections for Servitization Business Model

6.3.2 Business Model: One-Time Purchase

Under this one-time business model, healthcare facilities acquire the device via an upfront capital expenditure. Because hospitals are not charged a clinical service fee by PExA for individual usages, this financial framework

is inherently decoupled from the servitization model discussed previously. Instead, long-term commercial sustainability relies on mandatory equipment servicing, which establishes a reliable recurring revenue stream through annual maintenance agreements. Furthermore, owing to this independence from the number of tests, this commercial framework exhibits superior scalability of population, making it significantly more adaptable when conducting commercial assessments for international markets. Given that the foundational prerequisites of this commercial assessment broadly align with the previously analyzed servitization model, the subsequent sections will delineate only the distinctive economic variables unique to this direct-purchase configuration.

- **Service fee:** Primary data from developer interviews indicate that the current servicing fee is anchored at 4% of the initial hardware acquisition cost. However, the company's commercialization road map targets an increase of this rate to 10% upon the market launch of PExA 2.1. To account for this strategic shift within the financial feasibility study, a sensitivity analysis is formulated across three distinct operational projections: a 4% baseline scenario reflecting current agreements, a 10% optimistic scenario aligned with the commercial targets, and a 2% pessimistic scenario to evaluate potential price suppression in the highly competitive market.

As illustrated in 6.4, the net present value (NPV) evaluations, a fundamental shift in the underlying financial variables occurs when transitioning to a transactional One-Time Purchase business model. Specifically, operational variables inherent to the servitization model, such as fluctuating screening populations, rental penetration metrics, individual test pricing, and initial access fees, are eliminated from the forecasting framework. Conversely, recurring equipment maintenance rates are introduced as the primary determinants of long-term commercial sustainability. While all of the projections still present negative financial outlooks due to high initial R&D and manufacturing expenditures, this direct-acquisition configuration demonstrates different financial performance compared to the service-oriented alternative.

One-Time Purchase			
Variables:	Worse	Base	Best
Purchase Volume:	120	180	240
Manufacturing cost (Price/Unit):	0.056	0.0500	0.046
Product Price:	0.07	0.075	0.08
Discount rate	0.09	0.07	0.05
Development Cost:	8.4	6	3.6
Marketing and Support:	0.5	0.4	0.3
Equipment and Tooling:	0	0	0
Service income: (% of Price)	2.0	4.0	10.0
Service cost:	450	300	150
Net Present Value(Million USD)	-\$13.3	-\$7.9	-\$1.0

Figure 6.4: Sensitivity Analysis and Financial Projections for One-time Purchase Business Model

6.3.3 Business Models in National Screening

This section evaluates the operational and financial implications of integrating PExA 2.1 into a national screening framework. Under this top-down policy configuration, the most critical economic determinant shifts to the reachable and aware fraction. Benchmarked against mature Swedish public health initiatives, most notably national breast cancer screening programs, which historically maintain an average participation rate of approximately 81% [50]. Based on this empirical reference, the reachable and aware fraction is formulated across three distinct projection scenarios to test the model's robustness: a 70% pessimistic scenario, an 80% baseline scenario, and a 90% optimistic scenario. Simultaneously, the target demographic expands exclusively to the population aged 55–74,

increasing the absolute potential user base to approximately 2,000,000 individuals. When filtered through the scenario percentages mentioned in advance, this demographic foundation yields the adjusted screening populations detailed in the subsequent evaluations, ranging from 1,050,000 in the worst scenario to 1,762,200 in the best scenario.

Consequently, the total hardware procurement volume must scale proportionally to accommodate the expanded screening population under this national screening. From an operational capacity planning perspective, the required institutional fleet size is projected by dividing the total annual screening volume by the maximum annualized throughput capacity per device, accounting for standard single-day patient processing limits and the healthcare facilities' annual operating days.

As illustrated in the financial projections under Figure 6.5&6.6, a fundamental divergence in net present values occurs between the two models. Under the Servitization configuration, financial performance demonstrates a substantial surge across all scenarios, driven by the linear relationship between continuous clinical service fees and the scalable screening population. The NPV reaches 73.4 million USD even under the worst scenario, scaling to 590.2 million USD in the best scenario. Conversely, the direct-acquisition configuration under the One-Time Purchase model presents a limited growth potential. Because hardware sales volumes are inherently decoupled from ongoing clinical utilization, the transactional model, burdened by high upfront development and manufacturing expenditures, yields negative net present values in both the worst (-11.6 million USD) and base (-4.2 million USD) situations, achieving profitability only under the optimistic projection at 6.2 million USD.

Servitization			
Variables:	Worse	Base	Best
Screening Population (Person)	1050000	1360000	1762200
Rental Adoption(Unit)	365	472	612
Test Price (USD/patient)	50	100	150
Maintainance Cost(USD/unit)	450	300	150
Discount Rate	0.09	0.07	0.05
Product Cost(M USD):	0.056	0.0500	0.046
Development Cost (M USD):	8.4	6	3.6
Marketing and Support (M USD):	0.5	0.4	0.3
Initial Access Fee (M USD):	0.005	0.01	0.015
Equipment and Tooling:	0	0	0
Net Present Value(Million USD)	\$73.4	\$263.9	\$590.2

Figure 6.5: Financial Projections and NPV Analysis under the Servitization National Screening Plan

One-Time Purchase			
Variables:	Worse	Base	Best
Purchase Volume:	313	405	524
Manufacturing cost (Price/Unit):	0.056	0.0500	0.046
Product Price:	0.07	0.075	0.08
Discount rate	0.09	0.07	0.05
Development Cost:	8.4	6	3.6
Marketing and Support:	0.5	0.4	0.3
Equipment and Tooling:	0	0	0
Service income: (% of Price)	2.0	4.0	10.0
Service cost:	450	300	150
Net Present Value(Million USD)	-\$11.6	-\$4.2	\$6.2

Figure 6.6: Financial Projections and NPV Analysis under One Time Purchase National Screening Plan

6.4 Strategy and Risk

As evaluated in the preceding sections, PEXA 2.1 demonstrates significant competitive advantages upon its commercial launch. By offering a non-invasive solution for the early screening of lung cancer, PEXA 2.1 effectively addresses a critical gap within the current lung cancer diagnostic and treatment approaches. Furthermore, previous geographical and strategic analyses indicate that PEXA's operational location and structural characteristics provide distinct advantages for European market expansion and commercial profitability. To synthesize these strategic dynamics, a comprehensive SWOT analysis is detailed below:

6.4.0.1 SWOT Analysis

- **Strengths:** Compared to traditional Low-Dose Computed Tomography (LDCT), PEXA offers a significantly lower cost per test, enhancing its financial attractiveness to primary care clinics and regional health centers. Additionally, its small physical footprint maximizes adaptability within small-to-medium-sized healthcare facilities. Structurally, the capacity of PEXA 2.1 to bridge both academic research and commercial diagnostic markets allows the company to achieve high strategic flexibility.
- **Weaknesses:** A primary clinical limitation is that PEXA has not yet identified specific biomarkers, creating a diagnostic bottleneck that may constrain its initial commercial adoption rate in commercial market. Furthermore, relative to the pricing of a single test, the initial capital expenditure and manufacturing costs of the PEXA device remain high, thereby prolonging the payback period required to get original R&D and development expenditures back. Consequently, a critical internal challenge lies in optimizing production processes to reduce manufacturing costs while simultaneously lowering ongoing maintenance fees.
- **Opportunities:** Under the Servitization model, aligning recurring revenue streams directly with the screened population yields superior profitability, particularly if PEXA 2.1 is integrated into national screening frameworks. Externally, establishing a robust operational and regulatory foothold within Sweden serves as a strategic gateway, facilitating streamlined market access into broader European healthcare markets after launch.
- **Threats:** Financial projections reveal that market success is heavily dependent on external pricing dynamics, specifically the willingness-to-pay of healthcare providers and the maximum out-of-pocket expenditure accepted by patients, both of which introduce substantial baseline uncertainty. Additionally, whether PEXA can secure public healthcare subsidies from the government remains a critical point that will heavily dictate the initial market competitiveness.

Overall, PEXA possesses substantial strategic elasticity when navigating its commercial market entry, as both evaluated business models demonstrate robust profit potential depending on final pricing configurations. Ultimately,

reducing internal manufacturing costs and aligning with the public's maximum price acceptance are the two key factors deciding PExA's long-term profitability.

Chapter 7

Exploitation plan

This chapter outlines the strategic plan for the market rollout and long-term viability of PExA 2.1. Combining the technical development and the commercial models discussed in previous chapters, this plan focuses on transforming PExA from a research tool into a widely adopted medical solution. The following sections will detail the commercialization strategy, market and sales implementation, plans for continuous product development, and the long-term financing strategy to ensure sustainable growth.

7.1 Commercialization Strategy

The commercialization strategy for PExA 2.1 is based on a step-by-step market rollout. The core plan is to first establish the product among research organizations before expanding into the clinical medical market.

7.1.0.1 Phase 1: Research and Early Validation Market (Current Phase)

Currently, the main customers for PExA 2.1 are university laboratories, research institutes, and development departments in pharmaceutical companies. At this stage, PExA serves as an advanced tool that allows researchers to collect tiny lung particles through a completely non-invasive breathing process. By analyzing the information carried by these particles, researchers can study hidden risks of lung diseases. Taking lung cancer as the main example, the goal in this phase is to work with research partners to identify the specific protein patterns related to early-stage lung cancer. This phase focuses on building scientific trust and proof without facing strict medical regulations yet.

7.1.0.2 Phase 2: Medical Examination Market and Business Model Transition (Future Target)

Once specific lung cancer protein patterns are successfully found and verified, PExA will begin entering the regulated medical market. Before entering this market, however, PExA must pass a series of safety and quality standards set by government health authorities. After passing these regulations, PExA 2.1 will target medical institutions, including hospitals, specialized clinics, and health checkup centers. The actual users will be people at high risk for lung cancer (such as heavy smokers or people exposed to hazardous work environments). PExA 2.1 will fill a major gap in current lung checks. Right now, screening relies heavily on low-dose CT scans (LDCT). However, CT scans can only see structural changes, meaning they detect tumors that have already formed. PExA, on the other hand, can catch microscopic changes at a molecular level before any physical tumors appear, achieving true early detection. To enter this market smoothly, a commercial model will be implemented as described in Chapter 7, a system based on renting the equipment out and selling the single-use collection parts through a continuous subscription. This model lowers the upfront cost for hospitals, making it much easier and faster for them to adopt the device.

7.1.0.3 Phase 3: National Screening and Mass Market Expansion

As more data is collected on successful early diagnoses over time, the ultimate goal is to scale up the business. A plan to partner with national health authorities will go into action to include PExA lung cancer screening in national public health programs. By making it a standard, government-supported checkup, a steadily increase PExA 2.1's market share will take shape, while saving lives on a much larger scale.

7.2 Market and Sales Plan

To successfully bring PExA 2.1 into the commercial market, the market and sales plan focuses on three core pillars: defining clear value propositions for key stakeholders, improving operational efficiency, and building industry trust.

7.2.0.1 Value Propositions for Key Stakeholders

The sales strategy directly addresses the specific needs and requirements of three critical groups involved with the device:

- **Patients (The End-Users):** The primary selling point is patient comfort and safety. Unlike painful invasive checks, PExA 2.1 provides a completely non-invasive, quick, and reliable testing experience, which significantly increases patient willingness to undergo routine screenings.
- **Operators (Health Centers and Staff):** For clinic managers, PExA 2.1 is presented as a highly cost-effective investment that offers a high Return on Investment by introducing a new, high-demand testing service. For the medical staff, the device features an intuitive design that allows them to conduct tests with maximum time-efficiency and minimal training.
- **Service Personnel:** The product features a modular design that is easy to assemble and disassemble. This minimizes the time required for routine maintenance and repairs, ensuring maximum equipment uptime—a major benefit that guarantees continuous revenue for both PExA and the medical centers.

7.2.0.2 Market Validation and Feedback (Field Validation)

- **Market Feedback:** Conduct another round of semi-structured interviews with early clinical adopters and operators. This allows us to validate if their efficiency requirements are fully met and identify any shifting market needs before large-scale commercial rollout.
- **Testing Efficiency Optimization:** To solve the clinical pain point of long patient waiting times for screening result, the market entry strategy focuses on minimizing the time from particle collection (via the tray) to final lab analysis, ensuring an efficient, and low human error workflow.

7.2.0.3 Institutional Trust and Relationship Building

- **Regulatory Compliance:** By strictly complying with national medical regulations and achieving high quality standards, PExA 2.1 gains official regulatory admission. This regulatory compliance serves as a government-backed guarantee, giving health organizations the confidence to adopt the product.
- **Academic and Industry Visibility:** To build industry reputation prior to full commercial launch, PExA can actively publish clinical breakthroughs in lung cancer detection in top-tier medical journals, while showcasing PExA 2.1's capabilities at major medical device exhibitions and respiratory seminars.

- **Early Commercial Collaborations:** PExA can establish partnerships with key potential customers (such as leading health examination networks) to build long-term institutional trust and secure the future commercial success.

7.3 Continuous Product Development

To ensure the long-term success and continuous improvement of PExA 2.1, a structured development plan is established across three core dimensions:

7.3.0.1 Future Development Areas

Based on feedback from the prototype testing, several improvement spaces have been identified to better meet the requirements of patients and clinics:

- **Patient Ergonomics and Screening Flexibility:** To allow patients to adopt the most comfortable posture during testing, an aim to extend the external sampling tube is set. Future research will focus on thermal conduction and fluid dynamics within the extended tube to ensure that temperature changes do not cause protein particle loss or condensation.
- **Hardware Optimization and Cost Reduction:** The internal structure will be redesigned to make the components more compact and simplified. This will lower manufacturing costs, making PExA 2.1 a highly competitive and affordable product for medical centers.
- **Biomarker and Cost Control:** On the scientific front, R&D will continue to identify specific protein patterns directly linked to lung cancer while improving laboratory workflows to keep the final analysis cost within a reasonable range.

7.3.0.2 Team Structure

To support these development areas, PExA's internal organization will expand to include specialized teams:

- **Regulatory and Quality Assurance Team:** This new unit will be responsible for navigating medical device regulations in target markets. They will conduct safety benchmarks against competing products to ensure PExA meets the highest standards of reliability.
- **Cross-Functional R&D Team:** Combining software, hardware, and biomedical engineers to execute the continuous product updates described above.

7.3.0.3 Resource and Partnership

- **Academic and Technical Collaboration:** PExA will maintain its long-term partnership with Chalmers University of Technology and other relevant universities. Collaborating on mechanical, electrical, and biomedical engineering research will maximize R&D efficiency, provide access to advanced laboratories, and accelerate technology iterations at a lower cost.
- **Clinical Collaboration:** To raise industry awareness of PExA 2.1, PExA will form strategic partnerships with leading medical institutions, lung cancer research centers, and the related organizations. By collaborating with these trusted authorities on clinical studies and co-hosting medical seminars, PExA can leverage their professional reputation to introduce the technology to the wider medical community, building long-term trust and accelerating market adoption.

7.4 Financing Plan

The financial strategy for PExA 2.1 focuses on balancing its positive social impact on early lung cancer survival rates with a highly viable commercial model designed for long-term growth. In the early stages, development and validation costs could be minimized through collaborations with academic institutions such as Chalmers University of Technology, which allows PExA to spend less R&D expenses by accessing advanced university laboratories while gaining fresh, innovative ideas from joint research and talent networks. This approach probably combined with PExA's self-funded capital, initial backing from investors, and the revenues from the academic research market.

As the project gets into the mid-stage and enters the commercial medical market, primary funding will come from commercial sales to health examination centers, which can be further supported by government innovation subsidies, such as those from Sweden, aimed at accelerating clinical trials and local healthcare adoption [52]. Furthermore, during this mid-stage, PExA will lower the production cost per unit through scale manufacturing, leveraging larger production volumes to decrease component expenses and improve profit margins. Based on Chapter 7, PExA can adopt flexible commercial models, including servitization and one-time purchases, to adapt to various market environments and lower the purchasing barriers for clinics.

In the late stage, the goal is to integrate PExA into national health screening programs, securing substantial government subsidies and funding while generating high profit margins through the massive volume of routine testing, ensuring long-term financial sustainability. By leveraging the credibility of being adopted into a national screening program, PExA can more convincingly expand its technology into the markets in other countries and attract further opportunities from global investors, successfully creating a positive financial recycle for continuous growth.

Chapter 8

Discussion and Recommendations

8.1 Fulfilling requirements

To evaluate the final design outcome, the system was assessed against the initial engineering specification. The requirements have been categorized and color-coded based on their level of fulfillment: **Green** indicates full compliance, **Yellow** indicates partial fulfillment or pending full validation, and **Red** indicates requirements that were not met within the current development cycle.

Functional Requirements

- **Adjustable Mouthpiece Height:** The mouthpiece height shall be adjustable within a vertical range of 0–30 cm.
- **Sensory Feedback Recognition:** At least 90% of users shall correctly recognize and interpret the sensory feedback provided during operation.
- **Valve Operation:** The valve shall be operable with a single action and require a maximum force of 1.5 N.
- **Internal Component Accessibility:** Components requiring maintenance shall be accessible within no more than three disassembly steps.

Usability Requirements

- **Cleanability:** All external surfaces shall be cleanable within 2 minutes, following the defined cleaning procedure.

Technical Requirements

- **Volume:** The total system volume shall be less than 0.125 m³.
- **Assembly/Disassembly Durability:** The EMC solution shall remain fully functional after 15 assembly/disassembly cycles.
- **Production Cost:** Production cost does not surpass current cost.

Regulatory Requirements

- **Electrical Safety:** The system shall comply with IEC 60601-1.
- **Risk Management:** Risk management shall be conducted in accordance with ISO 14971.
- **Quality Management System:** Development and manufacturing processes shall comply with ISO 13485.
- **EU Medical Device Regulation Compliance:** The system shall comply with Regulation (EU) 2017/745.

The level of fulfillment across the different requirement categories reflects the specific scope and objectives of the current development phase.

Primary focus was directed toward Functional Requirements and Usability Requirements, as establishing a working proof of concept and ensuring intuitive user interaction were critical to verifying the core design principles. Consequently, the mechanisms regarding height adjustability, feedback integration, and valve operation were fully implemented and validated within the prototype architecture.

In contrast, Technical Requirements could only be partially verified. While volumetric targets were successfully achieved within the boundaries of the physical prototype, this parameter remains uncertain for the final commercial layout. Similarly, long-term operational metrics, such as component durability over extended assembly cycles, cannot be definitively proven until the design transitions from a functional prototype to a finalized manufacturing standard. Furthermore, the production cost requirement was not met, as the prototype proved more expensive than the existing machine due to the inclusion of the newly introduced motorized height adjustment sub-system. This cost increase was accepted because user interaction and usability were intentionally prioritized over cost constraints during this phase.

Finally, Regulatory Requirements were deliberately excluded from the evaluation scope at this stage. Because the current iteration serves as an early-stage development prototype rather than a finished, market-ready medical device, full compliance verification with strict international standards remains a downstream objective to be addressed during clinical trial preparation and industrial scale-up.

8.2 Continued development

During the project, several alternative technical solutions and design directions were identified that were not fully implemented in the final concept but remain relevant for future development.

One important discussion concerned the heating system for the long tube. The final concept focused on insulating the tube to maintain temperature stability during particle transport. However, an alternative solution using an actively heated tube was also considered. Although the heated solution could potentially simplify the overall system design and reduce dependency on insulation performance, it would likely require continuous maintenance and replacement over time due to wear and heating element degradation. In contrast, the selected insulation-based solution may offer better long-term robustness and lower maintenance requirements, despite being mechanically more complex.

Another discussion involved the valve control mechanism. Several approaches were explored, including manual lever control, touchscreen operation, and timed automatic control. Supporting multiple interaction methods increases flexibility and allows adaptation to different user preferences and clinical workflows. Different healthcare environments may also require different levels of operator involvement depending on staff experience, patient condition, and time constraints. However, introducing several parallel control methods may also increase interface complexity and reduce clarity for first-time users. In clinical environments, where procedures must often be performed quickly and consistently, too many interaction alternatives may create uncertainty regarding the correct workflow. This could increase cognitive load for both operators and patients, particularly during stressful situations or when the system is used infrequently. Another challenge is that multiple control systems may reduce overall system transparency. If different operators use different interaction methods, the procedure may become

less standardized, potentially affecting consistency between measurements and making troubleshooting more difficult. In addition, introducing more interaction options increases both software and hardware complexity, which may negatively affect reliability, maintenance requirements, and future regulatory validation. A more simplified interaction strategy could therefore improve usability, reduce training requirements, and create a more intuitive experience for first-time users. At the same time, reducing the number of available control methods would also reduce flexibility and adaptability for experienced operators or specialized clinical situations. This highlights a broader trade-off throughout the project between creating a highly flexible system and maintaining simplicity, clarity, and robustness in everyday clinical use.

The degree of automation within the system was also discussed extensively. The current prototype mainly demonstrates a visual and functional concept rather than a fully autonomous system. One future possibility would be to automate valve control using internal particle measurement data from the laser system. For example, the valve could automatically close when the detected particle concentration decreases below a defined threshold. However, this approach introduces technical uncertainty because the airflow dynamics within the reservoir system may delay or distort particle measurements. An alternative solution could involve using flow sensors positioned near the mouthpiece to detect patient breathing directly and synchronize valve operation in real time. While this could improve automation and reduce operator involvement, it would also increase system complexity and require additional validation to ensure reliability in clinical use.

This subject also opens up a broader discussion regarding patient experience and human interaction in healthcare environments. While increased automation could reduce operator workload and improve process consistency, it may also reduce transparency and make troubleshooting more difficult in clinical settings. At the same time, the role of healthcare personnel during the procedure raises important questions regarding patient comfort and perceived safety. Some patients may feel more secure when guided directly by healthcare staff throughout the breathing maneuver, especially since the procedure can initially appear unfamiliar or technically advanced. Human interaction may therefore reduce uncertainty and create a more reassuring testing experience. On the other hand, some patients may experience less stress if they can perform the procedure independently and at their own pace without feeling observed or pressured by medical personnel. A more autonomous process could potentially create a calmer and more comfortable environment for certain users. This highlights that the optimal level of automation may not only depend on technical performance and efficiency, but also on psychological and human-centered factors related to patient trust, comfort, and perceived control during the procedure. Because different patients may prefer different levels of assistance, a flexible semi-automated approach may ultimately provide the best balance between efficiency, usability, and patient experience.

Finally, the prototype revealed practical integration challenges related to the HEPA filter tube and height adjustment system. The current prototype configuration does not fully accommodate the filter tube within the adjustable structure, meaning that additional mounting solutions and increased adjustment height would likely be required in future iterations. This highlights how subsystem integration became an important challenge during development, where individually functional solutions still needed further refinement to operate effectively as a complete product system.

Overall, these discussions demonstrate that the development process involved continuous trade-offs between usability, technical complexity, flexibility, manufacturability, and long-term maintainability.

8.3 Concluding action plan

This action plan outlines the recommended steps for product commercialization, based on the report's analyses, evaluations and commercial assessment:

The first plan of action is to do more trials with people who aren't in perfect healthy condition. This would include people with bad backs who are having people using the current machine. The marketing personnel should start making people aware of the device to increase demand in preparation of release. The first physical change would be to install the screen. Because of the amount of data and test needed to make it accurate the development should start early.

Once the first plan has started the development team should start looking in to the fitment of the lead-screw mech-

anism. During this process it should be decided how the automated system should be done as it would be part of the redesign process and further look into how the internal components can be arranged to ease service.

Long-term development will focus on scaling operations beyond the initial Swedish market. Future commercial efforts should target geographic regions with higher concentrations of respiratory diseases, where the PExA device's diagnostic capabilities will yield a greater clinical and financial impact.

Chapter 9

Conclusion

The goal of the project was to develop redesign recommendations for the existing PExA 2.1 device for its public introduction. This was done by focusing on the user interactions with the device. The prototype addresses flaws in the current design and improves the patient's overall experience with using the machine.

The prototype allows for the evaluation of three user experiences. The closet closing mechanism facilitates faster and easier access to the inner components for service personnel, the lead-screw mechanism enables easy adjustment to each patient's height, and the screen assists the patient and operator with the complex breathing maneuver. Together, these features create a more efficient, accommodating, and self-explanatory workflow for the patient, nurse, and technician alike. For future development and implementation of the tested sub-functions, the current PExA 2.1 would need to be redesigned with consideration of the components of the new functions. The system would need to be fully automated for the tested screen to work as intended, and the internal components rearranged with the closet doors in mind. Consequently, the next steps must focus on integrating these separate sub-functions into a single, cohesive, and manufacturable product architecture.

In summary, this project successfully delivers strategic design recommendations to prepare the PExA platform for clinical environments. By transforming critical user touchpoints into functional engineering requirements, this new architecture directly resolves existing ergonomic flaws. In doing so, these advancements lay the groundwork for a standardized, user-centric device that is fully optimized for widespread adoption across standard healthcare centers.

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Appendices

Reflection Report

This reflection report discusses and examines the design methodologies applied during the redesign development of the PExA device project, based on the framework presented by Ulrich et al. (2020) and the course lectures. The aim of this report is to reflect on the practical relevance, strengths, and limitations of the methods chosen and implemented throughout the project, rather than only presenting the final results.

To provide a clear chronological overview, this reflection follows the actual product development timeline, beginning with prestudies and requirements, followed by early exploration, concept generation, concept evaluation and selection, and finally prototyping and physical verification. Methods that were considered but later rejected are integrated and discussed within their respective development phases to provide a comparative reflection on the decision making process.

1 Prestudies and Requirement Specifications

1.1 Product Requirement Specification

In the project, the method of establishing product requirement specifications was utilized. According to Ulrich et al. (2020), product specifications describe the product in terms of measurable metrics and target values, which guide the development process. While various qualitative research techniques, such as clinical observations, stakeholder interviews, and public surveys, are excellent ways to gather customer needs and help us understand what users want, these identified needs alone do not offer sufficient technical direction for an engineering team. Translating subjective user preferences into measurable metrics and target values is a critical methodology for steering the development process and avoiding purely opinion based decisions. Product specifications do not tell the team exactly how to solve the problem, but they create a clear and shared understanding of what the product should achieve.

The process starts by creating a list of metrics based on customer needs. Then, benchmarking is used to compare existing products and understand what performance levels are realistic. After that, target values are set, often including both ideal values and minimum acceptable values. Later in the process, the specifications are refined. This includes making tradeoffs between different requirements, since improving one aspect may negatively affect another. The specifications are also used as a guide throughout the development process and can be updated when new information appears (Ulrich et al., 2020). In the case, the customer survey and benchmarking were carried out in a previous course. A choice was made to reflect on and utilize these insights so the product could start directly with creating the correct specifications, which then they were used as a guide throughout the development phase.

This method was chosen because it helped make the project more structured and clear. By defining measurable requirements early, it became easier for the team to understand what the team were aiming for and to stay aligned. Another reason is that the method supports better decision making. Instead of relying only on opinions, the team could compare different ideas based on how well they met the specifications. This made it easier to evaluate concepts and justify the choices objectively. It also helped us in the early ideation phase to generate ideas that actually met the requirements in the specification. The method was also very useful for connecting customer needs to technical solutions, ensuring the final product addressed what users want rather than what the team assumed.

However, a major limitation of this method is that it was difficult to give exact numbers to subjective human experiences. For example, turning something like patient comfort into a fixed numerical target sometimes felt too simplified early in the process. In addition, following the early specifications too strictly could create a rigid process that limited creative problem solving. If the team is unwilling to update the specifications as new insights are discovered, the document can become a barrier to innovation instead of a helpful guide. Overall, while the method reduced uncertainty and provided clear direction, the team learned that it must be used with flexibility.

2 Early Concept Exploration

2.1 Experimental and Engineering Prototyping

Although participating in the course workshop was a mandatory requirement where the team had no choice but to build physical models, it proved to be a highly beneficial methodological step for the project. The team engaged in what Ulrich et al. (2020) define as experimental or engineering prototyping. These models were never intended to represent the final clinical device, but rather were used mainly for rapid physical exploration to test ideas and enable early learning. Experimental prototypes are typically developed quickly using simple materials and focus on specific functions or ideas instead of the entire product. According to Ulrich et al. (2020), their primary purpose is often to support team communication and test hypotheses in order to identify strengths and weaknesses, effectively reducing early project uncertainty before committing resources to a potentially flawed concept. During the workshop, the team used this method to visualize and evaluate several early concepts in a practical way. Instead of only discussing ideas theoretically, the team could test them directly and compare different solutions. This exercise helped us quickly identify which ideas had potential and which concepts were less suitable for further development.

One key advantage of this approach was that it encouraged fast iteration. Because the prototypes were simple and inexpensive to build, the team could experiment with multiple ideas without investing too much time or resources. The prototypes also improved communication within the team, since physical representations made it easier to explain concepts and receive feedback compared to only using sketches or verbal discussions. However, relying on this early prototyping method also had some limitations. Because the prototypes were very simplified, they could not fully represent how the final product would function in reality. Some ideas that initially appeared promising during prototyping later proved less practical when considering technical constraints and usability in more detail. In addition, the short workshop format limited how deeply the team could test each concept and also the material the team used was limited. Overall, even though the mandatory workshop had clear limits in time and materials, using early experimental prototypes was very valuable for the project. It supported the idea generation, improved team communication, and helped us identify weak concepts. This reduced technical uncertainty before the team moved into the more structured and detailed development phases.

3 Concept Generation

3.1 Concept Classification Tree (Means Function tree)

In the project, the team used the means function tree, which Ulrich et al. (2020) refers to as a concept classification tree, as a method to structure and explore different solution concepts during the ideation phase. The primary purpose of the method is to organize ideas into categories and subcategories, making it easier to understand the design space and identify possible combinations. According to Ulrich et al. (2020), this tree structure is used to break down a problem into different dimensions and classify possible solutions within each category. Each branch of the tree represents a different approach or solution principle, providing a visual overview of alternative concepts. This helps a design team move from unstructured brainstorming to a much more systematic exploration of solutions.

At the beginning of the concept generation phase, the team created a function list using verb noun descriptions, such as "support weight" or "transfer force." This step helped us identify the main function, subfunctions, and desired functions of the respiratory device. By defining functions in this highly structured way, the team could clarify the complex overall problem and break it down into smaller, manageable subproblems. This approach made it much easier to continue working with the means function tree, since the clearly defined functions provided a strong foundation for exploring different solution principles systematically. Instead of randomly generating ideas, the team could identify key aspects of the problem and develop targeted alternatives within each category.

the team chose to use the concept classification tree not only to generate new ideas, but primarily to understand how the existing PExA product works and how its different internal parts are connected. the team used the method

to analyze and structure the current solution by linking known functions to existing solution principles. This helped us break down the physical product into its main function and subfunctions, allowing us to understand how each specific function is fulfilled in the current design. By visualizing this architecture in a structured way, it became much easier for an external observer or the entire team to see how the different parts of the product relate to each other.

Reflecting on this process, a major reason for using this method was that it drastically improved the foundational understanding before the team started concept development. By clearly mapping functions and solutions, the team created a better foundation for later ideation. This significantly reduced the risk of misunderstanding the core mechanical requirements and helped ensure that any new concepts would directly address the key functions. The method also made it much easier to identify potential areas for improvement; by analyzing how each function was currently solved, the team could clearly see where alternative, more efficient solutions might be possible.

However, using this method also had some drawbacks. Breaking a complex clinical device into very detailed subfunctions can sometimes lead to fragmented thinking. The team noticed that by focusing too much on categorising and improving individual parts, it was easy to lose track of how these elements would later come together in the overall product design. In addition, building a full classification tree takes a lot of time, and it sometimes felt like simple mechanical actions were being made more complicated just to fit the strict structure of the method. Even with these limits, the method was still very useful overall because it helped us better understand the existing product, clarified the link between functions and possible solutions, and gave us a clear and structured base for further development work.

3.2 Rejected Method: The Process Flow (Black Box) Model

the team also considered and attempted to use the process flow model referred to by Ulrich et al. (2020) as the "Black Box" model. This method aims to link customer needs and early solution ideas by focusing on what a system does, defining clear inputs and outputs, without focusing on how the system works internally. In theory, treating a complex clinical device as an abstract process should have made early ideation easier. The team tried to map the machine's operation using this framework, defining the main input as uncollected respiratory biomarker particles and the final output as securely collected diagnostic samples.

However, when the team reflected on the results of this exercise, the team found that the method did not fit the engineering context and was therefore not used further. Since the main diagnostic technology of the PExA machine is patented and outside the redesign scope, the black box approach did not give us any useful new information. It only repeated general inputs and outputs that the team already understood, and it did not help us analyse or improve the specific modular parts of the product that the team could actually change. In addition, the high internal complexity of the device, together with limited time and resources, meant that simplifying its operation into a general workflow made it harder to understand the real mechanical details the team needed to work with. The model became too abstract and did not support deeper technical understanding or decision making. Compared to the concept classification tree, which helped us map and study detailed internal mechanical relationships, the black box model stayed too general and did not add real value within the project limits. This showed an important lesson: high level abstraction tools do not work well when a project requires focused redesign of specific and constrained internal parts.

3.3 Information Gathering (External and Internal Search)

For the generation of subsolutions for each subfunction, ideas were derived from both external and internal searches. As emphasized by Ulrich et al. (2020), implementing an existing solution is almost always faster and more cost effective than inventing a completely new one from scratch. A conventional solution to one specific problem can often be combined with a highly novel solution to another, resulting in a superior overall design. Ulrich et al. (2020) suggest initially gathering information broadly without committing to any particular direction. This involves looking across entirely different industries, technologies, and application areas, even those that seem only loosely connected to the clinical device problem. The primary goal is to avoid missing potentially useful solutions by being too narrow in scope too early in the project. Once a broad range of promising solutions is discovered, the team narrows its focus and investigates the most relevant subsolutions in depth.

For external search, the team gathered insights from lead users, including health center personnel and test patients. Health center personnel contributed subsolutions related to the "Provide Guidance" and "Height Adjustment" subfunctions, while test patients provided input across all subfunctions in the morphological matrix. In addition, the team benchmarked the device against related products in both similar and entirely different markets that share relevant subfunctions. For instance, different types of CPAP machines were researched to generate subsolutions for the "Transfer Breath" subfunction.

On the other hand, internal search is about using the team's own knowledge and creativity to generate new solution concepts. Ulrich et al. (2020) describe this as the most open ended and creative activity in the entire product development process. Unlike an external search, which brings in existing solutions from outside sources, internal search creates new ideas from knowledge already present within the team. The team arranged dedicated group meetings to generate ideas for each subfunction in a shared environment. During these sessions, the judgment of ideas was strictly suspended, as Ulrich et al. (2020) suggest resisting the urge to reflect critically during the pure generation phase. Instead, the team channeled any judgmental tendencies into suggestions for tangible improvements or alternative concepts. While this dual approach was highly effective for gathering a large volume of ideas, a notable negative aspect of external searching is that it can sometimes anchor the team's creativity too heavily to existing competitor designs, subtly limiting the imagination of what is possible.

3.4 First Morphological Matrix Concept and Idea Generation

The concept combination table, widely known in design methodology literature as a morphological matrix, is a systematic method used during the concept generation phase to explore and organize combinations of subsolutions across different subfunctions. As described by Ulrich et al. (2020), it forms a critical part of a structured concept generation approach. The table is organized so that each column represents a subfunction derived from the prior functional decomposition, which in the case was the means function tree. For the project, the first iteration of this matrix was intentionally kept highly focused, including only the subfunctions deemed most impactful and feasible within available timeframe. This decision was also reinforced by the suggestive guideline from the book to focus on subfunctions whose solutions depended on each other and eliminate independent subfunctions, whose subsolutions can be chosen regardless of the other columns.

This resulted in six subfunctions: "Adjust Height", "Transfer Breath", "Provide Guidance", "Control Airflow", "Control Valve", and "Communicate Data". Each column was then filled with alternative subsolutions, gathered through external and internal search methods. Once all functional columns were filled, individual concept generation took place. Rather than working exclusively as a group, each team member was given extended time to independently reflect and generate ideas. This independent approach is strongly supported by Ulrich et al. (2020), who note that individuals working alone tend to produce a higher quantity and quality of concepts than when working collectively in a group setting. This independent session resulted in numerous concepts of varying form and complexity.

However, reflecting on this first iteration, a major limitation was that it was overly simplified. By artificially limiting the matrix to save time, the resulting concepts functioned theoretically but severely lacked the technical depth required for a complex clinical tool.

4 Concept Selection and Iterative Development

4.1 Concept Screening with Elimination matrix

Moving into the evaluation phase, after the first idea generation, the team had 13 design concepts that needed to be narrowed down efficiently. As highlighted in the Systematic Design lectures, an Elimination Matrix is an important step in evaluating concepts to remove disqualified solutions, ensuring the team only keep alternatives that actually solve the main problem. To do this, the team built our elimination criteria directly from our most important requirement specifications. Each concept was evaluated to see if it fulfilled these demands using four clear symbols: (+) for Yes, (-) for No, (?) when more info was needed, and (!) if the team needed to check with

the specification. Concepts that received the highest number of negative (-) scores were eliminated from further development.

The elimination matrix was very useful for reducing the number of concepts in a structured way, especially during an early phase with limited time and information. It removed obvious failures without requiring complex mathematical weighting. However, the method also showed clear limitations in our project. Since many concepts were still underdeveloped, several evaluations became highly uncertain and depended heavily on team assumptions rather than verified data. As a result, the matrix risked eliminating ideas too early based on our own incomplete understanding rather than actual technical feasibility.

4.2 Second Morphological Matrix and Iterative Refinement

After the initial Elimination Matrix, four design concepts were removed, leaving nine alternatives. Although reduced, the number of remaining concepts was still relatively large. At this stage, it also became clear that the initial concept generation, based on the first Morphological Matrix, was overly simplified and did not include all the functions required to develop fully detailed design concepts. The first matrix had been reduced to limit the total number of generated concepts, but this decision resulted in designs that lacked important technical detail and functional completeness.

To address this limitation, a second Morphological Matrix was developed based on the project's means function tree, providing a more complete representation of the product. This revised matrix included both critical requirements and desired features, rather than focusing only on the aspects that were easier to modify during development. Following the Elimination Matrix, this more detailed approach was applied by having each team member systematically go through all functions and combine different solution options. This process helped us further develop the remaining concepts and led to six well developed and comprehensive design concepts.

Our approach to a full product development concept was surprisingly similar to the "Distort Ideas" technique described in the textbook, as it involves modifying or rearranging elements of existing solutions to generate new ones. It also aligns with the SCAMPER method, a framework that guides this creative process through seven structured prompts: Substitute, Combine, Adapt, Modify (or Magnify/Minimize), Put to other uses, Eliminate, and Reverse or Rearrange (Ulrich et al., 2020).

While this second matrix successfully added necessary depth, its primary limitation was that it generated a very large number of possible technical combinations. While this theoretically increased our creative output, it also made the process incredibly difficult to manage and evaluate systematically. Many combinations became highly unrealistic or overly complex, meaning the team still had to rely heavily on subjective judgment when selecting which combinations were actually worth pursuing.

4.3 Further Idea Developments

With the second morphological matrix as support, and by widening the idea generation to consider the product as a whole rather than only focusing on individual functions, each group member was assigned to further develop concept ideas independently as homework with a set deadline. Each individually developed concept was then presented to the group, which reflects the "use of related stimuli" method, where each person first creates their own ideas independently. This approach ensures independent thinking, avoids early group bias, encourages diverse perspectives, and leads to more developed and varied ideas.

Using this independent idea development strategy, the concepts were also presented in different ways. Some concepts were developed using CAD to provide a more immersive 3D understanding. Detailed sketches were also used to highlight different functionalities of the product concept, including both internal and external features. In a format highly similar to the Gallery Method described by Ulrich et al. (2020), a formal concept presentation session was held. During this structured session, team members reviewed each concept, listened to the creator's explanations and internal thought process, and provided spontaneous feedback. Reflecting on this phase, working separately on different concepts was very effective for generating detailed clinical designs, but it also created a risk that team members became too attached to their own ideas. This made the later objective scoring phases slightly

more tense than expected.

4.4 Concept Scoring with Pugh and Kesselring Matrices

To support the final concept selection, the team used Pugh and Kesselring matrices as structured decision making tools. The Pugh Matrix was done twice with different reference concepts, and helped organize discussions and compare concepts against a common reference solution. This created a clearer overview of strengths and weaknesses between alternatives and made it easier to narrow down the remaining concepts systematically (Ulrich et al., 2020).

However, the method also had limitations. Since several concepts were still relatively undeveloped, some evaluations had to be based on assumptions rather than verified technical data. In addition, the scoring process remained partly subjective, since team members interpreted certain criteria differently. The outcome was also influenced by the selected reference concept, meaning that another baseline could potentially have changed the relative evaluation of the concepts.

After the qualitative screening, the remaining concepts were evaluated using a Kesselring Matrix. Compared to the Pugh Matrix, this method allowed a more quantitative comparison by assigning weights to different criteria based on project priorities such as usability, manufacturability, and operational efficiency. The matrix helped create a more structured final discussion and clarified which concepts achieved the best balance between technical feasibility and user requirements (Ulrich et al., 2020). At the same time, the Kesselring Matrix also involved subjective decisions, particularly when assigning weights to criteria. Although the numerical scoring created a sense of objectivity, the final ranking still depended heavily on how the team prioritized different requirements. Small changes in weighting could therefore influence the final result significantly. Overall, the combination of the two methods was useful because the Pugh Matrix supported early qualitative screening, while the Kesselring Matrix provided a more detailed comparison of the strongest remaining concepts. Together, they helped structure decision making, even though engineering judgment and group discussion still played an important role in the final selection.

4.5 Rejected Method: Prototype and Test (for Concept Selection)

Ulrich et al. (2020) describe "Prototype and Test" methodologies as a very effective approach for concept selection, where teams build physical functional models of different concepts and choose the best option based entirely on test results. It was considered that this method for our selection phase but decided not to use it. Our only experience with building physical models came from a single early workshop, where the team quickly explored our first ideas. When working through the Elimination Matrix, the team realized that the team did not have enough information for several of the concepts, especially regarding whether the proposed long tube solutions would actually work as intended.

Building fully functional and detailed models to test exact physical behaviors for multiple complex clinical concepts would have been entirely unrealistic. The internal structure of a PEXA machine is highly complex and such testing would require a large amount of time. The extra effort needed to design and build test setups for each option would also have delayed the project schedule. Because of these severe resource and time constraints, the team chose not to use physical testing at this stage. Instead, a structured decision tools such as the Pugh and Kesselring matrices to evaluate and compare the concepts based on theoretical reasoning and expected performance were used.

4.6 Rejected Method: User Concept Testing

Ulrich et al. (2020) also detail concept testing as a common method used to evaluate how customers respond to early designs before an organization moves forward with final development. This process usually involves presenting ideas to a representative group of users to collect information about their preferences, using anything from verbal descriptions and 3D CAD renderings to fully working prototypes. Although this is a very practical and valuable

method in gathering consumer product information, the team decided to reject it due to the specific needs of our end users.

The clinical device is meant for a very narrow target audience, primarily specialized healthcare professionals and patients within specific risk groups. This makes it very difficult to gather a sufficiently large and relevant statistical sample within the project timeframe especially with different public and private healthcare centers. Furthermore, asking users to test only small parts of a design makes it hard for them to understand the whole experience, especially without fully working models to show them. Instead, the team relied on Pugh and Kesselring matrices, supported by well defined product specifications to represent user needs from our previous customer needs report as a guideline.

5 Conclusion

In conclusion, the use of structured design methods played an important role in our project. Methods such as product specifications, prototyping, concept classification trees, and selection tools like the Pugh and Kesselring matrices helped us move from unclear ideas to a more structured and refined final concept. One of the main strengths of these methods was that they supported our choices and improved communication within the team. By using clear criteria and structured approaches, the team were able to reduce project uncertainty and justify our complex design choices in a more objective way. The methods also helped us explore a wider range of ideas before narrowing down to the most suitable solutions.

However, this project has clearly shown that design methods are only tools, not fixed solutions. They must be used critically and adapted continuously to each specific situation. Matrices can contain hidden biases, early physical models can mislead design directions, and strict frameworks can limit innovation. Overall, the project has shown that design methods are valuable tools, but they must be used critically and adapted to the situation. Their usefulness depends not only on the method itself, but also on how and when it is applied.

Reference

Ulrich, K. T., Eppinger, S. D., Yang, M. C. (2020). Product design and development (7th ed.). McGraw Hill Education.

A Requirement Specification

REQUIREMENT SPECIFICATION									
				Issued by: Project group PEVA					
Category	Requirement	R/W	Importance	Values	Confidence with the Value (1-5)	Justification (why it is a requirement)	Reference (where the value come from)	Evaluation/Verification	
Functional Requirements	Adjustable Mouthpiece Height The mouthpiece height shall be adjustable within a vertical range of 0-30 cm.	R		0-30 cm	4	To ensure users of all sizes can sit comfortably and reach the mouthpiece easily.	European Anthropometric Data (ISO 7250), DIN EN Anthropometric Database (https://dined.o.tu-delft.nl/en/database/tool) (ISO 7250 data) with https://cdn.standards.iti.lt/altsamples/54419d7fb3b611453460419435983a2888f/ISO-11064-4-2013.pdf	1. Field testing with Extreme Users (Height) 2. refer the products on the market.	
	Airflow Resistance Reduction The airflow resistance through the mouthpiece and filter system shall be at least 15% lower compared to the reference design								
	Intuitive User Guidance At least 90% of users shall correctly recognize and interpret the breathing maneuver instruction guidance provided during operation	W	10	15 %	1	To reduce the effort needed for a patient to breathe through the filter system.	Observation, performance benchmarking between current and previous models		Testing with flow sensors (Check for feasibility)
	Valve Operation The valve shall be operable with a single action and require a maximum force of 6 N	R		>90%	4	To ensure most users can finish every step correctly on their own.	FDA (2016) https://www.fda.gov/media/80481/download : AAMI HE75	Field test with Users	
	Proper Installation Indication The system shall provide a clear mouthpiece installation indicator that is correctly recognized by the operator	R		≤ 6 N	3	To ensure the valve is easy to operate with a single action without straining the hand.	PEVA Design guidelines	Designers	
	Internal Component Accessibility Internal components shall be accessible for maintenance within 5 disassembly steps	W	4	>90%	4	To ensure most operator scan install the mouthpiece correctly and safely without help.	FDA (2016) https://www.fda.gov/media/80481/download : AAMI HE75	Field test with Users	
	Leakage Prevention The system shall prevent air leakage under normal operating conditions.	R		within 5 steps	3	To minimize downtime and ensure easy maintenance by service personnel.	PEVA Serviceability Design Guidelines	Service Personnel testing	
Usability Requirements	Breathing Pause Capability The system shall allow the patient to pause between breathing cycles for 5-10 seconds without affecting system functionality or safety	W	11	5-10 sec	1	Sometime the machine leaks which is not sufficient	Project team	Design Engineers	
	Learnability A first-time user shall be able to understand and complete the full operating procedure within 1-3 minutes without external assistance	W	5	1-3 min	3	Easier to understand the procedure	Observation of patient breathing patterns during usability testing and consultation with healthcare staff.	Field testing with Users	
	Mouthpiece Comfort The mouthpiece shall enable users to comfortably maintain it in the mouth for at least 30% longer duration compared to the previous design	W	8	30%	2	To ensure comfort during extended periods of use	Compared the automatic BP monitor which usually take people 1-3 min to know the whole process flow.	Field test with Users	
	Mobility The system shall be movable by one person without mechanical lifting assistance	W	1	1 person	3	To ensure the system can be moved safely by one person between 'health center rooms.	Manual handling https://cdn.standards.iti.lt/altsamples/768207d0ddad006e29445538a3d9c648201a5/ISO-11228-1-2021.pdf	Trial with Customers	
	Cleanability All external surfaces shall be cleanable within 5 minutes, following the defined cleaning procedure	R		5 mins	2	To ensure the device is safely decontaminated between users in a short timeframe.	https://mdcop.com/doc/standard/ISO17864-2-2021(E).pdf Hospital Hygiene Protocols	Test in lab	
	Assembly Time The system shall be assembled within 5 minutes, using step-by-step assembly instructions.	W	7	5 mins	2	In order to stream line the assembly and minimize lead times	PEVA assembly testing	Service Personnel testing	
	Volume The total system volume shall be less than 0.125 m³	R		< 0.125m³	5	Make the machine smaller	PEVA Design guidelines	Design engineers	
Technical Requirements	Weight The total system weight shall not exceed 30 Kg.	W	2	< 30 kg	5	PEVA Design guidelines. To keep the device within the safe lifting limits for a single person.	Manual handling https://cdn.standards.iti.lt/altsamples/768207d0ddad006e29445538a3d9c648201a5/ISO-11228-1-2021.pdf	bench pressing the machine	
	Assembly/Disassembly Durability The EMC solution shall remain fully functional after 15 assembly/disassembly cycles	R		15 cycles	2	To improve sustainability and reduce waste caused by single-use adhesives/tapes.	PEVA Lifecycle and Sustainability Analysis	Test in lab	
	Production Cost Production Cost doesn't surpass current cost	R		\$75.000		Keep production cost low	PEVA Internal Information	Bill of Materials	
	Serviceability The system shall be serviceable using standard commercially available tools.	W	6	screw driver		No need for specialized tools		Design Engineers	
	Warm-up Time The system warm-up time shall not exceed 30 minutes.	W	3	< 30 min	4	To maximize the number of patients screened per day, the screening time must be minimized. The most critical factor in reducing total screening time is decreasing the heating time.	PEVA internal technical specifications	1. compare the time difference between before/after 2. make sure all of components work normally	
	Electrical Safety The system shall comply with IEC 60601-1	R		IEC 60601-1	4	Electrical safety requirements for medical electrical equipment, EU regulatory expectations for CE marking of medical devices and harmonized standards under Regulation (EU) 2017/745	IEC guidelines : https://mdcop.com/doc/materialDownload/IEC%2060601-1-2%202020.pdf		
	Risk Management Risk management shall be conducted in accordance with ISO 14971	R		ISO 14971	4	ISO 14971 to identify and control risks	https://mdcop.com/doc/standard/ISO14971-2019(E).pdf		
Regulatory Requirements	Quality Management System Development and manufacturing processes shall comply with ISO 13485	R		ISO 13485	4	Compliance with ISO 13485 ensures controlled processes, regulatory compliance, and patient safety.	https://mdcop.com/doc/materialDownload/ISO%2013485%202016%20Medical%20devices%20Part%202%20Guide.pdf		
	EU Medical Device Regulation Compliance The system shall comply with Regulation (EU) 2017/745	R		(EU) 2017/745 (MDR)	4	The system shall comply with Regulation (EU) 2017/745 (MDR) to meet legal requirements for medical devices in the EU	https://health.ec.europa.eu/medical-devices-topics-interest-theme_en https://www.cs.mcgill.ca/~mcf/ra/ra/actualites/normes-normes/see-sh-four-mars-2024-392-harmonisation	Pass all Medical tests	

B Stakeholder analysis - tree diagram

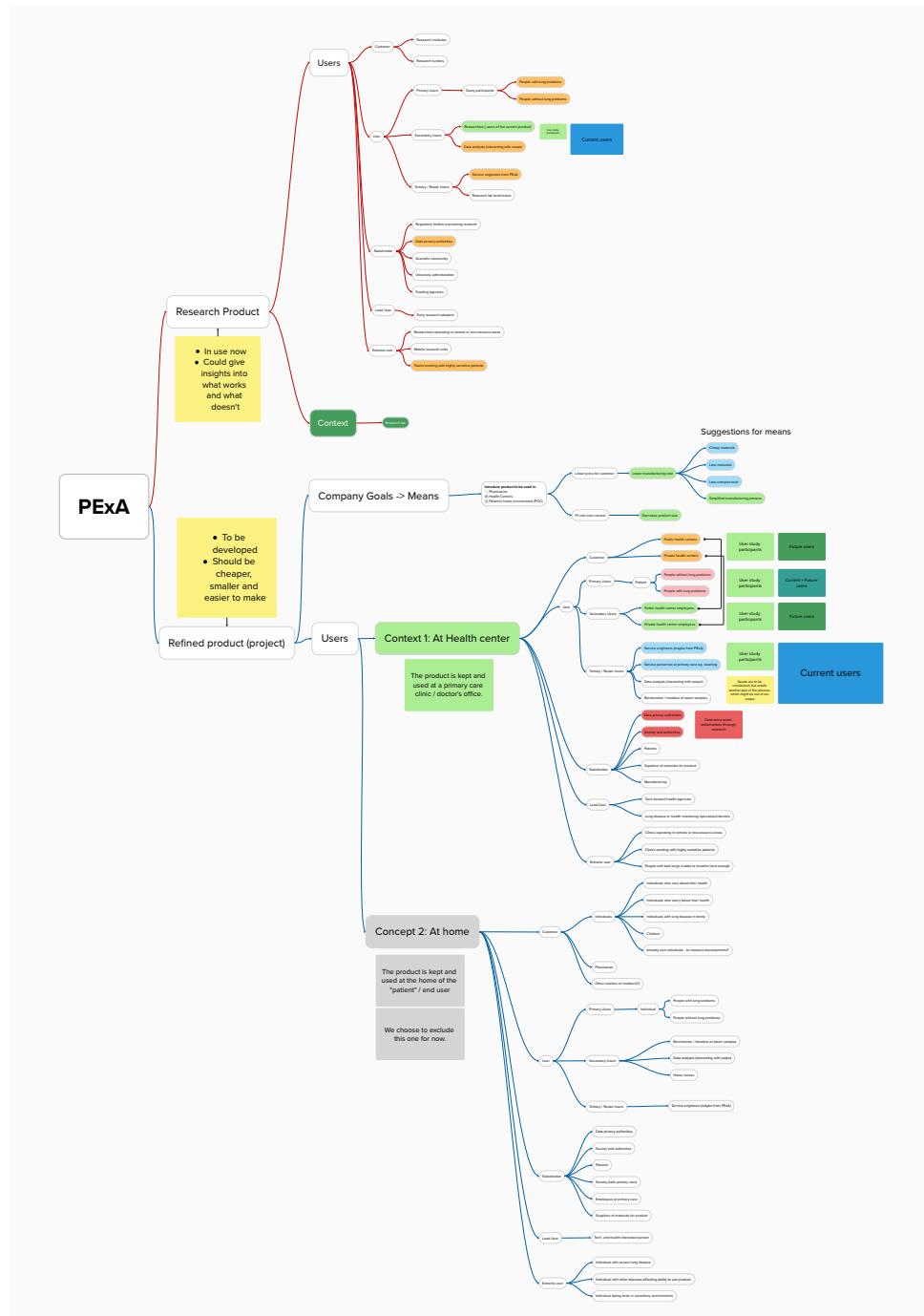


Figure B.1: Stakeholder Analysis Tree Diagram

C Hierarchical Task Analysis - tree diagrams

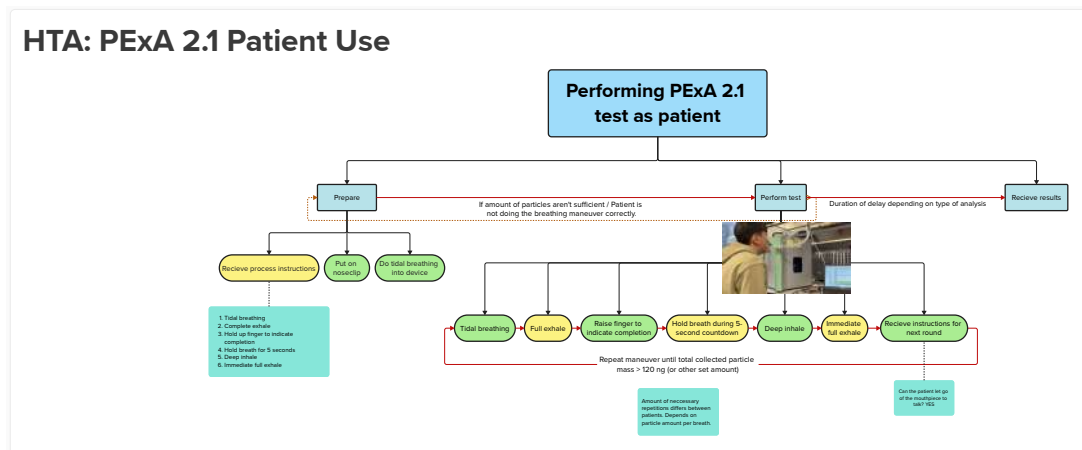


Figure C.1: HTA for patient using PEXA 2.1

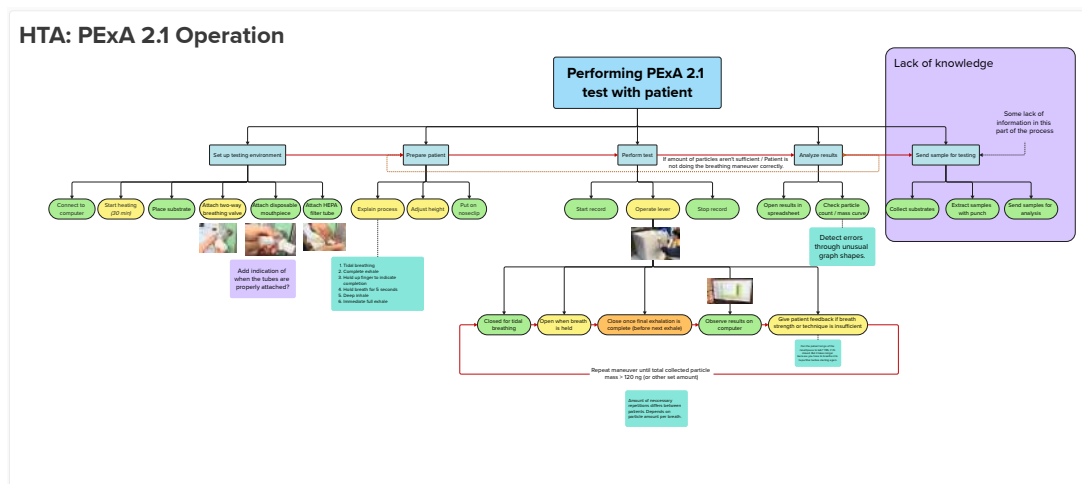


Figure C.2: HTA for operating PEXA 2.1

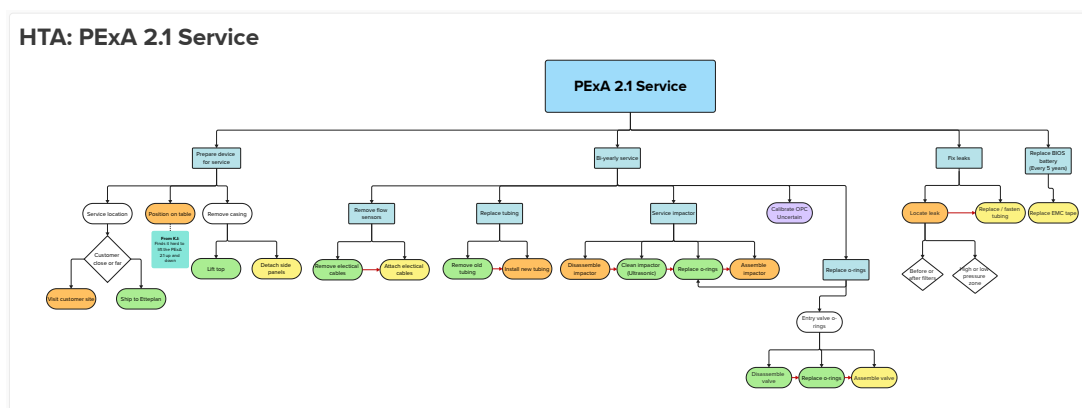


Figure C.3: HTA for service personnel servicing PEXA 2.1

D KJ Method Clusters

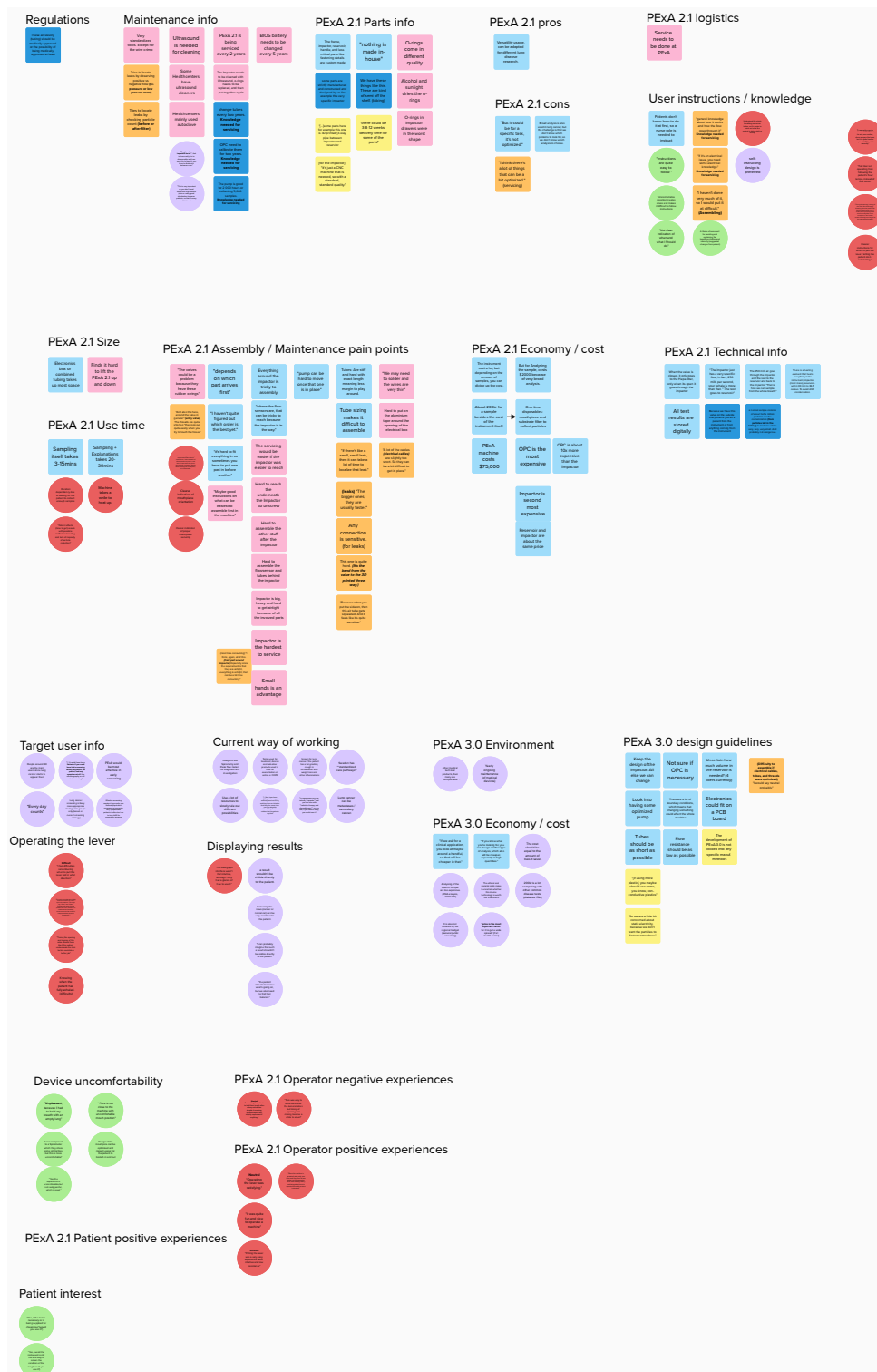


Figure D.1: Results from first iteration of KJ Method

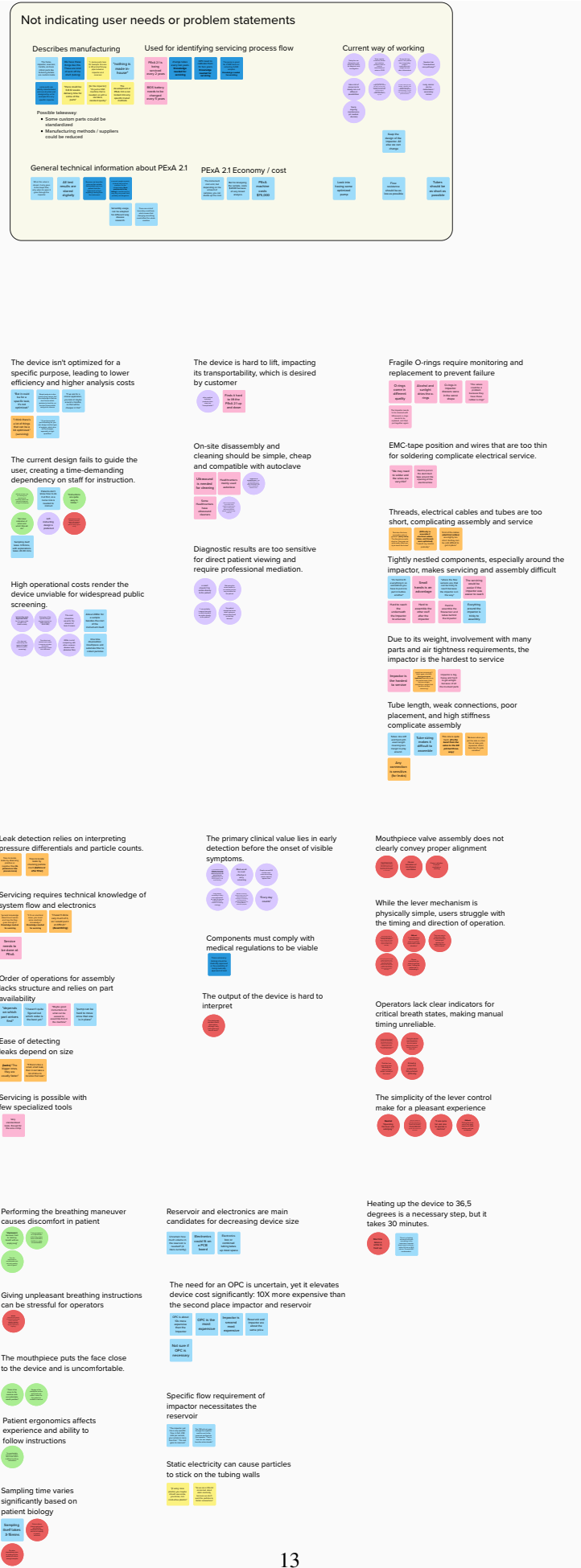


Figure D.2: Results from second iteration of KJ Method